Knowledge
Confidence
Skill to take exams

Mark Klimek Lecture Notes

Systems go down w/ low pH
↑ pot

LECTURE 1: Acid Base Balance & Ventilator

Interpreting blood gases

(remember the rules of the B's)

systems go up w/ high pH
↓ pot

- If the pH and the bicarb are both in the same direction then it's metaBolic (Bicarb Both Bolic), if they are in different directions then it is respiratory
- If bicarb is normal and the pH is low or high then its respiratory
- You will be given 8 values for arterial blood gas, always first look at the pH and the bicarb first
- You get acidosis and alkalosis from the pH

LABS: ABG's
The normal pH is 7.35-7.45
The normal bicarb is 22-26 (the bicarb years where you make all the decisions [22-26 years old], or 2+2+2=6)
The normal CO2 is 35-45 (same as pH)

Signs and Symptoms with ABG's

- As the pH goes up so does my patient
  - If the pH goes up, every system in your body gets more irritable/hyperexcitable
- As the pH goes down so does my patient
  - If the pH goes down, systems in your body shut down
- Except for potassium- When pH goes down, potassium goes up
- **If the pH goes up (alkalosis):** you will find irritability, hyperreflexia (3&4), tachypnea, tachycardia, borborygmi (increased bowel sounds), seizure (need suctioning at the bed side because they can seize and aspirate)

+2 for reflexes are normal

- **If pH goes down (acidosis):** hyporeflexia, bradycardia, lethargy, obtunded, paralytic ileus, coma, respiratory arrest (need bag-mask ventilation bag at bedside for respiratory arrest), +1 reflexes
- **MACkussmal**- compensatory and respiratory pattern for only acid base disorder: MAC- Metabolic ACidosis

Respiratory Acidosis multiple choice example: What would you see with a patient who is in respiratory acidosis?

a. **+1 reflex,**
b. **diarrhea,**
c. **adynamic ileus (no movement),**
d. **spasm,**
e. **urinary retention,**
f. **paraxysmol atrial tachycardia,**
g. **second degree lovitz, type 2 heart block (impulse is being slowed),**
h. **hypokalemia**

LAB: REFLEXES
0&1-hyporeflexia
2-normal
3&4- hyperreflexia

EXAMPLE: (In general what do pain meds do?
ANSWER: They sedate you, they are CNS depressants: lethargy, lucidity, reflexes at +1, hyporeflexia, obtundent

Causes of Acid Base Imbalance What will cause the acid-base imbalance

- **Don't get signs and symptoms mixed up with causation!!!**
- What causes something is the opposite of what the signs and symptoms are
  - EXAMPLE: diarrhea will cause a metabolic acidosis but once you get acidotic, it will shut your bowels down and you will get a paralytic ileus.

- The first question you should ask yourself if the scenario involves a lung problem.
  - Is it a respiratory problem? BUT remember it can still be respiratory acidosis/alkalosis…
- Next question you ask yourself…
  - is the client overventilating or underventilating?

  - If the patient is **overventilating** pick **alkalosis**
  - If they are **underventilating** pick **acidosis**

- If the client is **overventilating**.. it has an attachment to the word- **alkalosis** (because they are both OVER)… **ventilating OVER becomes respiratory ALKALOSIS**

- If the client is **undeventilating**.. it has an attachment to the word- **acidosis** (because they are both UNDER)- **ventilating UNDER becomes respiratory ACIDOSIS**

Examples:

pick metabolic alkalosis.. Why?

- Pt is losing acid... pt will become basic

  Otherwise everything else that is not lung or the above, pick metabolic acidosis

Ex.

1) Patient had GI surgery and has had an NG tube to low intermittent gone post suctioning for 3 days, what acid base disorder would he most likely exhibit?
   - Metabolic alkalosis
2) Patient has hyper emesis gravidarum , what acid base disorder are they going to exhibit
   - Metabolic alkalosis
3) Continuation: Pt is going to be dehydrated- what acid base disorder would they have?
   - Metabolic acidosis
4) Pt has acute renal failure, what acid base disorder would this be?
   - Metabolic acidosis- its not lung or vomiting or suctioning so it has to be metabolic acidosis
5) A pt with infantile diarrhea would have what acid base disorder?
   - Metabolic acidosis
6) A pt with third degree burns over 60 percent of the body?
   - First phase- metabolic acidosis

no vomit
no suctioning

If you don't know what it is, just choose metabolic acidosis!!

default setting.

RECAP

What do you have to know for Acid Base?

- If the **pH** and the **Bicarb** are **both** in the same direction, its **metabolic**
- The **direction** my **pH** goes, **so does my patient**, except for potassium
- MACkussmal- compensatory and respiratory pattern for only acid base disorder: MAC- Metabolic Acidosis
- Overventilate: (alkalosis) - translate the word
- Underventalate: (acidosis) - (translate the word)

1) A woman is overzealously using her breathing techniques during labor, what acid base disorder will she exhibit? **Overventilation**

   - Respiratory **Alkalosis**

2) A child is near drowning, what acid base disorder would it be? **Underventilating**

   - Respiratory Acidosis

3) Your patient has emphysema, what acid base disorder would it be? **Underventilating**

   - Respiratory Acidosis

Ventilating does not mean respiratory rate.. respiratory rate is irrelevant- ventilation has to do with gas exchange!!

Examples:

1) Patient has pneumonia in 4 lobes of the lung, breathing at 50/min and their SO2 is at 78 on 8 liters per max

↓ low O_2
↓ pH

   - Explanation: Breathing really fast while still having a low O2 level means that the patient is still **underventilating** because respiratory rate has nothing to do with it. Everyone pays so much attention to rate when they should be paying closer attention to the SO2.

   - If your SO2 is good and you are breathing slow, you are fine but if your SO2 is low and you're breathing fast, you are actually **underventilating.** A lot of times the respiratory rate compensates- pay attention to SO2!!!

2) Patient is on a PCA pump, what acid base imbalance would tell you they need to come off that thing?

↓ decreased respirations
↓ pH

   - A PCA pump depresses respirations. So, patients need to come off of it as soon as possible because if they were getting too much it would make their respiratory rate go really down which would make the patient underventilate so the answer would be **respiratory acidosis.**
   - So **respiratory acidosis** would tell you that you need to come off the PCA pump.

What if its not lung?
It would be Metabolic.

Only one scenario that you will answer metabolic alkalosis: if the patient has prolonged gastric vomiting or suctioning

D. Begin to decrease the settings

ANSWER:
C. Hold the order and call the doctor: because pt underventilating on the ventilator and without it he'd be even worse. If he has respiratory alkalosis it could mean he's being overventalated, which means he doesn't need the machine.

Lecture 2: Alcoholism, Delirium Tremors & Peak and Trough

Alcoholism: Psych + Med surg

Psychodynamics- The **number one problem** of psychological in alcoholism is the same exact problem for any abuse: **DENIAL**

- Abusers have an infinite capacity to deny
- Denying allows the abuser to keep doing it without having to answer for it
- It is number 1 because how can you treat someone who denies they have a problem and until they admit they have a problem.
- **Definition of denial**: refusal to accept the reality of a problem

How do you treat denial?

- Confronting it by pointing out the difference to the person of what they say and what they do
  - "okay, you say you are not an alcoholic but its 10 am and you already drank a 6 pack"
  - "You say you're not a child abuser but protective services has your children"
- confrontation (attacks the problem) is not the same as aggression (attacks the person)
- Don't ever choose answer that uses the word **YOU** with confrontations only the letter **I**

Deny --> confront

- Not with loss and grieving (DABDA- denial, anger, bargaining, depression, acceptance)

What do you do for the denial of loss and grief?

- Support it because it serves a function

Dependency and Codependency

- Dependency- when the abuser gets the significant other to do things for them or make decisions for them (the abuser is dependent)

- Codependency- when the significant other derives positive self-esteem from making decisions for or doing things for the abuser
  - Pathologic and yet symbiotic relationship
  - Abuser gets a life without responsibility and a SO gets positive self-esteem

Treatment?

- Set limits and enforce it- teach SO to start saying **NO** and to keep doing it

- Must work on self-esteem of codependent person first or it will never work- the dependent abuser is going to make them feel bad when they start saying NO and emotionally manipulate them

- Codependent person has to say.. I AM saying NO because I'm a good person- (usually the relationship will break up in the end)

Manipulation

- abuser gets SO to do things for him or her that is not in the best interest of the significant other.
- Nature of the act is interest and harmful

- This is like dependency because in both situations the abuser is getting the SO to do things for them and you can tell the difference by Neutral vs. Negative

- If what the significant other is being asked to do is **inherently harmful or dangerous** to the SO it is **manipulation**

- If what the significant other is being asked to do is neutral, **NO HARM NO FOUL**- it is **dependency and codependency**

Examples

1) A 49 yr old alcoholic gets her 17 yr old daughter to go to the store and buy alcohol for her
   - manipulation because minor buying alcohol is illegal

2)A 49 yr old alcoholic asks her 50 yr old husband to go to the store and buy alcohol for her.
   - Dependency because there is no harm

Treatment for manipulation-

- Set limits and enforce them- start saying NO
- It is easier to treat than dependency and codependency because nobody likes being manipulated and there is no positive self-esteem issue

How many pts do you have with denial if bob is in denial? 1
If bob is dependent how many patients do you have? 2 (dependent + codependent)
If Bob is the manipulator how many patients do you have? 1

ALCOHOLISM

- Wernicke's-Korsakoff syndrome
    - Wernicke's encephalopathy
    - Korsakoff psychosis

But they tend to go together because you find them in the same patient

Wernicke's korsakoff is

1) Psychosis induced by vitamin B1 or thiamine deficiency – you lose touch with reality and go insane (loss of touch with reality) because you don't have B1
2) Amnesia (memory loss) with confabulation (making up stories)- make up stories because they forget. They are psychotic because they believe it. Lie is just as real as reality and their memory loss is "what happened to the 90's and they'll lose entire decades of their memories". Will often have an entire psychotic reality- real as anything else that is happening

How would a nurse deal with this?
Bad: confronting is wrong because it's due to brain damage and most likely permanent
Good: redirect the patient- take what the patient can/cant do and channel it to something that he can do "well why don't you go shower so we can watch what the news of the day Washington D.C. TV" if he's insisting he's part of government. **DO NOT present reality because they cannot learn it.**

Characteristics: take B1 to prevent

- **Preventable:** take Vitamin B1- coenzyme necessary for the metabolism of alcohol so if you don't have B1 you will not metabolize alcohol and you will not go into KREB cycle where it would get used up for energy so it will instead accumulate and go into the brain and destroy brain cells

- **Arrestable:** which means you can stop it from getting worse by giving B1, stopping drinking is not necessary

- **Irreversible:** 70 % irreversible (go with the majority, 2 good news one bad news)

DRUGS that have to do with alcohol

Antabuse (revia)- **DISULFURAM**

1) What is it used for?
    - Aversion therapy - want alcoholics to develop a gut hatred for alcohol- when you take this drug it will interact with the alcohol level in your blood and make you horribly ill – to the point you couldn't even pay them to drink- works in theory better than it works in reality – doesn't work as well as it says it does

2)What is the onset and duration of its effectiveness-

- 2 weeks 2 weeks before they can drink safely on and off the drug. Usually Dr. will prescribe pill, then pt is taken to a transition home for 2 weeks to assure that they take the drug and then let out to the community where every time they drink they will get deathly ill but if they decide they want to want to drink at a (lets say) high school reunion, pt will need to stop taking it two weeks before.

Teach these patients to avoid all forms of alcohol to avoid nausea, vomiting and possibly death

3) What do they need to avoid? all forms of alcohol read everything all labels.

- need to avoid mouthwash and aftershaves because they will get sick- nauseated
- perfumes and colognes, insect repellents, any over the counter that ends in the word ELIXIR (all have alcohol), alcohol based hand sanitizers, no unbaked icings (all have vanilla extract which will make them sick)
- **DO NOT** pick the red wine vinaigrette if it's on a multiple choice question because it has no alcohol and they can have it

OVERDOSES AND WITHDRAWS

Every abused drug is either an **UPPER** or a **DOWNER**- because they are the only drugs that do anything

Most abused type of drug that's not an upper or a downer is **laxatives in the elderly**

When you get an overdose question the first question you should ask yourself is…

1) **is the drug an upper or a downer?**

UPPERS

- Caffeine
- Cocaine
- PCP/LSD (psychedelic hallucinogens)
- Methamphetamines (crystal meth)
- Adderall- ADD drug

Signs and symptoms:

- Things go UP-
- Examples: euphoria, tachycardia, restlessness, irritability, borborgimi, diarrhea, 3&4, spastic, seizure (suction bag needed)

DOWNERS

Everything that is not an upper is a downer

Patients looking for an in between effect people will take both together

135 all together

Examples:

- Dilaudid
- Morphine sulfate

- Codeine
- Demerol
- Fentanyl
- Ambien
- Ativan
- Xanax
- Valium
- Librium
- Phenobarbital

Downners make you go down because they are downers

- lethargic- big danger is respiratory depression leading to respiratory arrest

Example question:

1) PT is high on cocaine, what is most important to assess?
    - check reflexes – because it is an UPPER

After you know whether the drug is an upper or a downer what is the second thing you ask yourself?

2) Are they talking about **overdose** (too much) or **withdrawal** (not enough)- because they are opposites

RECAP QUESTIONS TO ASK YOURSELF

1) is the drug an upper or downer
2) is it overdose or withdrawal

Overdose or intoxication- too much

RECAP:

- Overdose on an upper- everything goes up
- Downer and intoxication- makes everything go down
- Withdrawal- not enough/too little- **too little upper** makes everything go **down** and **too little downer** makes **everything go up**
- Upper overdose looks like what other situation: downer withdrawal
- Downer overdose looks like - upper withdrawal

What two situations would respiratory arrest and depression be the highest priority?

- Downer overdose and upper withdrawal

Which two would seizure be your biggest risk?

- Upper overdose and downer withdrawal

Example

1) Squad calls you about pt who has overdosed on cocaine what would you expect to see?

- upper
- overdose-too much upper
- CNS drug not autonomic drug
- **Seizure**
- **3-4 reflexes**
- **irritability**
- **increased temp**

2) You are caring for a client withdrawing from cocaine, what is expected?

- respirations less than 12 and difficult to arouse
- need narcan

DRUGs in the Newborn

- Always assume intoxication not withdrawal at birth
- After 24 hours assume baby is in withdrawal

Example:

You are caring for an infant who has qualine (downer) in system because of drug addict mom. What symptoms would you expect 24 hours after birth? SATA

a. **difficult to console**
b. ow core temp
c. **exaggerated startle reflex**
d. respiratory depression
e. **seizure risks**

AWS always happens first but DTS won't always follow.

Alcohol withdrawal syndrome vs. delirium tremors

a) every alcoholic goes through withdrawal 24 hours after they stop drinking
b) only a minority get delirium tremors after 72 hours
c) alcohol withdrawal syndrome always precedes delirium tremors, but delirium tremors do always follow alcohol withdrawal syndrome
d) AWS is not life threatening while DTS can kill
e) AWS are not a danger to self and others
f) DTS are a danger to self and others- unstable you can die
g) AWS- loud and obnoxious- because they are withdrawing from a downer which makes everything go up
h) Need to keep an eye on DTs because they are a danger to themselves and others

Differences between AWS and DT's

AWS

- Follow a regular diet

- Semi private anywhere on the unit
- Up ad lib- can go around anywhere they want to go
- NO restraints because they are not a danger to self or others
- Alcohol withdrawal patient can even be on overflow in Peds

DTs

- NPO or clear liquids because pt will be at risk for seizures (withdrawing from a downer makes everything go up) pt will get aspirations
- Private room near nurses station- dangerous and unstable
- Probably should be in ICU but not good for the rest of clients
- Nurse needs to decrease workload to take on DT's patient
- Restricted bed rest
- Must be restrained because they are dangerous, no bathroom, just bed pans
- Need to be in a vest or 2 point locked leather restraints (arm and a leg, opposite) and rotate it every two hours. Lock left arm and then right leg and then opposite

Both

- both get antihypertensive (both withdrawing from downers so everything is going up)
- Need a tranquilizer (both withdrawing from downers so everything is going up)
- Both get B1- (to prevent wernicke's and korsakoffs) "no b1, or you'll be 1"

DRUGS

- psych most common tested drug
- Insulin is 2nd most common
- Anticoagulants are 3rd most common
- digitalis 4th most common
- aminoglycosides 5th most common
- steroids 6th most common
- calcium channel blockers 7th most common
- Beta blockers 8th most common
- pain meds 9th most common
- OB 10th most common

AMNIOGLYCOSIDES- powerful class of antibiotics mean old drugs

- When nothing else works pull out the aminoglycosides gram - infections
- Dangerous
- Considered the gun for infections

AMINOGLYCOSIDES

- Think: "**a mean old mycin**" – that tells you they are antibiotics used to treat serious, life threatening, resistant, gram negative infections
- Treat a mean old infection with a mean old mycin
- Not for sinusitis or otitis media, or strep throat (not considered a mean old infection)
- But yes for tuberculosis, septic peritonitis, pulmonary pyelonephritis, septic shock, burn wounds 80% of body (SERIOUS INFECTIONS)

***Mycins- all end in mycin but not all drugs that end in mycin are aminoglycosides

3 mycins that are not mean old mycins are: THRO!

- Erythromycin
- Zi~~p~~thromycin
- Clari~~p~~thromycin

Ends in mycin- it's a mean old mycin. If it has thro, throw it off the list and only use for infections that are not that bad

What are the toxic effects?

1) MICE- MYCIN- Mickey Mouse Ears – Ototoxic (ear), monitor hearing, ringing in the ears (tinnitus), and vertigo or dizziness (ear has equilibrium)
2) Human ear- connect the dots and its' shaped like the kidney- nephrotoxicity- MUST monitor **CREATININE- the best indicator of kidney or renal function-** 24 hour creatnine clearance is better than serum creatnine.
3) **HAVE A VISUAL OF THE NUMBER 8-** the number 8 drawn inside the ear reminds you of the fact that they are.. Toxic to cranial nerve number 8 which is the EAR Nerve and you…
4) Administer them every **8** hours

What is the route?

IM or IV, do not give PO because they will not be absorbed

Oral mycin= goes into gut, dissolves and go right through you has no systemic effect

***** EXCEPT in two cases for oral!**

1) **Hepatic encephalopathy** or **hepatic coma**- when ammonia levels get too high and you go into a coma… you can die. Treatment is to get the ammonia down and **oral mycins do that** because it will kill gram negative bacteria in your gut, sterilize your bowel, **kill ecoli in your gut which is the number 1 producer of ammonia** in your gut, and decreases it. Because these people have liver damage we do not want it going to the liver and its perfect because it goes in and right out the gut. *Makes you have diarrhea that makes you get rid of stuff.
2) Give during preop bowel surgery- to **sterilize the bowel,**

******will not have otoxicity because it is not absorbed,**
Both: sterilized bowel without causing ototoxicity because it is not absorbed

What mycins are used for bowel sterilizers?

Neomycin and Kanamycin are typically used as bowel sterilizers

*****Sargent asks "who can sterilize my bowel?" "Neo Kan!"**

Trough and Peak

Trough- when the drug is at its lowest
Peak- when the drug is at is highest

mycin need TAPS

TAP levels ← way to remember the way to do it.
T-draw your trough
A- administer
P-Draw your peak

***Narrow therapeutic window- small window between what works and what kills
TAPS is important for this!

Example:

- Furosemide (Lasix)- wide range- TAPs not necessary
- Dig- .125-.25??- narrow therapeutic window- TAPS necessary

When do you draw trough? med doesn't matter route matters
Route matters-
Sublingual- 30 minutes before the next dose
IV- 30 minutes before the next dose — IV push
IM- 30 minutes before the next dose
Subq- 30 minutes before the next dose
PO- 30 minutes before the next dose

When do you draw the peak?
Depends on the route
***The same drug given in two different routes at the same time will have different peaks however two different drugs given at the same time and at the same route will peak together
Sublingual- 5-10 minutes after the drug is dissolved
IV-15-30 minutes after the drug is finished (when the bag is empty)
IM- 30-60 minutes after you give it
SUbq- SEE- See diabetes lecture (insulins)
PO- never test PO's because they're all too variable

Example:

1) 100 ml of a drug at 200 ml per hour- 30 min. hang it at 10, it'll finish at 1030 and you will draw the peak at 11. WHENEVR YOU GET TWO IN THE CORRECT RANGE PLAY THE PRICE IS RIGHT-THE HIGHEST WITHOUT GOING OVER (given 1045 as a choice also)

- Vomiting or suctioning=metabolic alkalosis
- Everything else is metabolic acidosis if I don't know what it is

always pay attention to the modifying phrase than the original noun example

- Person with OCD who is now psychotic… what is more important? Obpsychotic and not OCD

VENTILATION

Alarms and how blood gases articulate with ventilates

↑high alarm no air in

- **High pressure alarm** is set off **by increased resistance to air flow** (machine is having to push too hard to get the air into the lungs) – machine will set off a high pressure alarm (set alarms for appropriate pressures)

- What would cause the high pressure alarm? **OBSTRUCTIONS**
  1) Kinks in the tubing (unkink the tube)
  2) Water condensing within the tube/dependent loops (empty water out of tubing)
  3) Mucus secretions in the airway (change position, turn, cough, deep breath and if that doesn't work… THEN you suction)
     - **Nurses must only suction patients as necessary and only when you have already turned, cough and helped patient deep breath**

good example for drag and drop question

↓low alarm no air out

Low pressure alarms- decreased resistance- (too easy to push breath in) two disconnections. **DISCONNECTION**

1) main tubing (reconnect)
2) oxygen sensor tubing (senses the FIO2 right at the trache area- wire black coated.. goes right alone the tubing and comes right to the trache and hooks into the hole into the tubing)

Acid Base disorders

Respiratory alkalosis : Overventilating means ventilator settings may be **too high**
Respiratory acidosis: Underventilating means the settings are **too low**

Example

1) Dr. says to wean pt off vent in AM- 6 am ABGs show respiratory acidosis, what would you do?
   A. Follow the order
   B. Call respiratory therapy (**never pick answer where you don't do something and someone else has to do something**)
   C. Hold the order and call the doctor

Lecture 3: Calcium Channel Blockers, Arrhythmias, Chest tubes, Congenital heart defects

Calcium Channel Blockers – calms you down

- calcium channel blockers are like valium for your heart
- calms your heart
- given when heart is tachycardic, pt is having tacharrhythmias, had a heart attack and need to rest heart
- never give to stimulate heart
- negative inotropic, chronotropic and dromotropic- its like valium for your heart- relax your heart and calm it down

Positive Chrontropes:

- strengthen
- speed up and stimulate the heart
- they are stimulants

Negatives

- cardiac depressant
- weaken
- slow down and depress the heart

What do Calcium channel Blockers they treat?

A) Antihypertensives- relax your heart blood vessels and blood pressure goes down
AA) Antiangina- relax your heart, uses less oxygen, and decreases oxygen demand because it relaxes the heart- worst thing that can happen to person with angina is if their heart speeds up so we want to slow it down
AAA) Anti Atrial Arrhythmias- (will not treat ventricular tachycardia) treats atrial flutter, atrial fibrillation, premature atrial contractions
TRICK: supraventricular tachycardia- supra means above, and the atria is above the ventricle.

Side effects?

**think H&H
H) headache – vessels dilate in the brain causing migraine
H) hypotension- relaxes heart and blood vessels
*headache is great choice for SATA most times

NAMES of Calcium Channel Blockers:

- Anything ending in dipine

Example: Amlodipine

- Think "Dipin in the calcium channel" (NOT PINE BECAUSE MANY DRUGS END IN PINE BUT CALCIUM CHANNEL BLOCKERS ALWAYS HAVE **DIPIN in the word**)

Calcium channels you MUST KNOW BY NAME
VERAPAMIL ← VND
CARDIZEM - Continuous IV Drip Nifi

***Vital signs needed to measured before giving a calcium channel blocker, ex. Blood pressure because pt will be at risk for hypotension

Parameters for Calcium Channel Blockers:

- hold the calcium channel blocker if the systolic is under 100, must monitor the blood pressure continuously while on cardizem drip and if it was 98/52- slow down the drip and measure BP again to keep the systolic over 100

Cardiac Arrhythmias

- Know how to interpret rhythm strips
- Know 4 by site:
  - Normal sinus rhythm – p wave, qrs and t wave for every single complex, and peaks of QRS complex are evenly spaced
  - V fib- chaotic squiggly line, no pattern
  - Ventricular tachycardia – sharp peaks and jaggers with pattern – Big Saw
  - Asystole -down and out, crash and burn time, flat line

Important terminology

- QRS depolarization- ventricular, rule out anything that says atrial
- P wave- refers to anything atrial and rule out ventricular
- **6 rhythms most tested on NCLEX**
  - **Lack of QRS's- no QRS**- asystole (flat line)
  - **Form of atrial**- set P wave
  - **Saw tooth- flutter-** "I saw the teeth and my heart fluttered"
  - **Chaotic-** fibrillation
  - **Bizzare** always applies to **tachycardia** ventri
  - **Periodic bizzare wide QRS**- pvc- SALVO of PVC's =short run of Vtach, dr.'s don't care as much for these, low priority

PVC's are low priority but if there are more than **6 PVC's** in a **minute or in a row,** OR **if PVC's fall on the t wave of the previous beat (R on T phenomena) they become moderate but are never high-**
PVC's are good after an MI or a heart attack because that means they are reprofusing.

Lethal arrhythmias

- High priority

Lethal and kill you in 8 minutes or less and are super high priority

1) Asystole
2) V-fib

Both have no cardiac output which means no brain perfusion=dead in 8 minutes

Potentially life threatening... but not life threatening… making it high priority

1) ventricular tachycardia

What is the difference between Vtach and asystole?

- Vtachers have a cardiac output (Dr. will ask… is there a pulse with that?)

Treatment

1) PVC's and Vtach-both ventricular

- For ventricular use LIDOCAINE (lidocaine is not used in a lot of squads now in the bigger cities and they are using **Amiodarone** instead)- lidocaine is cheaper and has longer shelf life.

2) **Supraventricular arrhythmias – atrials – use the ABCD's-**
   a. **Adenocarb (adenosine)- need to push in less than seconds, fast IV push- usually when you don't know you go slow, but this drug must be slammed in less than 8 seconds with a 20 ml of IV fluid flush right after- but risk going into asystole for about 30 seconds but they can come out of it-**
   b. **Beta blockers- LOL- best class of drugs ever, negative chromo, negative drono, negative, valium for your heart, so they will treat A, AA, AAA- anti atrial arrhythmia- headache and hypertension as a side effect like calcium channel blockers (better for people with asthma because beta blockers vasoconstrict),**
   c. **Calcium Channel blockers- like valium for the heart, negative chromo, dromo & inotropic, treat A, AA, AAA with the side effect of H&H,**
   d. **Digitalis (digoxin)- LAMOXIN,**
3) **VFIB- U D FIB- shock them**
4) **Asystole- epinephrine and atropine**

CHEST TUBES

Purpose: to reestablish negative pressure in the pleural space so that the lung expands when the chest wall moves

Pleural space- where negative is good

Negative space- makes things stick together
Inside of the chest wall is a lining called the **parietal pleura**
Outside of the chest wall is a lining called the **visceral pleura** and it lines the outside of the lung between the parietal pleural and the pleural space
Good air exchange happens because there is **negative pressure in the pleural space**

Air and blood create **positive pressure** that **pulls apart** the lung and the chest wall- **creating no air exchange**

Chest tubes reestablish negative pressure in the pleural space so that the **lung expands when the chest wall moves**

In a **pneumothorax** the chest tube removes **air to reestablish negative pressure**
-air caused the positive pressure so a chest tube needs to go into remove the air and reestablish negative space in the pleural space

In a **Hemothorax,** the chest tube removes blood to reestablish negative pressure
-blood is causing the positive pressure and it needs to be removed to reestablish negative pressure in the pleural space

In a **pneumohemothroax,** the chest tube removes blood and air to reestablish negative pressure- blood and air is causing the positive pressure it needs to be removed to reestablish negative pressure in the pleural space

If boards gives you a question that says you have a client in with chest tubes for a hemothorax- what would you report to nurse, lpns, dr or rn?

1) chest tube is not bubbling
2) the chest tube drained 800 ml in first 10 hours
3) ~~the chest tube is not draining~~
4) the chest tube is intermittently bubbling

Disease for which the tube is placed

What does a hemo chest tube do?

- Drain blood so number 3 would need to be reported

What would you report in a pneumothorax,?

- The chest tube is not bubbling (needs bubbling)
- The chest tube drained 800 ml in the first 10 hours (doing something its not supposed to do)

***must pay attention to the disease for which it was placed to know what to expect

Location of the tubes
1) Apical- chest tube is way up high, which means you are removing air because air rises
2) Basilar- bottom of lungs and remove blood because blood is subject to gravity

Example:
Your apical chest tube is draining 300 ml per hour- BAD

Your basilar is draining 200 ml per hour – Good
Your apical tube is bubbling- GOOD
Your basilar tube is not bubbling – GOOD

A hemo would neeed a basilar tube
A pneumo would need an apical tube
And a hemo pneumo would need one of each

Example:
How many chest tubes, and where would you place a unilateral pneumo-hemothorax?
-2 chest tubes on apical for pneumo and basilar for the hemo
How many chest tubes and where would you place a bilateral pneumothorax?
-2, on apical

How many chest tubes and where would you place them for post op chest surgery?
-2, an apical and a basilar on the side of the surgery because you are to assume that chest surgery or trauma is unilateral unless otherwise specified

Trick question: How many chest tubes would you need and where would you place them for a post op, right pneumonectomy
-none because it's the removal of the lung and no pleural space

USED for
-lobectomy
-wedge resections
-etc.

TROUBLESHOOTING
What do you do if you knock over chest tube drainage water seal apparatus?
-set it back up and have patient take deep breaths- NOT A MEDICAL EMERGENCY

What do you do if water seal breaks?
-different because positive pressure can get into the pleural space so you
1) CLAMPIT first so nothing gets in ,
2) Cut it away from broken device
3) Stick end of tube in sterile water
4) Unclamp it because you reestablished water seal

*CLAMP, CUT, SUBMERGE, UNCLAMP
**its better to be under water than to be clamped because air cant go in but stuff can come out – water seal solves the problem

What is the **first** thing your going to do when the water seal breaks?
-Clamp (ORDER)
What is the **best** thing to do when the water seal breaks?
-Submerge tube under sterile water

BEST QUESTION IS DIFFERENENT THAN A FIRST QUESTION

Example:
1) You notice a pt has ventricular fibrillation on the monitor (no cardiac output + no pulse) what is the **first** thing you are going to do?

a) Place a backboard
b) Begin chest compressions — best

order question

This question is about order so you need to PLACE A BACKBOARD FIRST because it doesn't make sense the other way around

If you get the same question with the word best you would pick BEGIN CHEST COMPRESSIONS-because its technically the only thing you can do

2) What do you do if the chest tube gets pulled out?
   a) First thing- take a gloved hand and cover the hole
   b) Best thing- cover with vaseline gauze

Where is it bubbling and when is it bubbling- ask yourself this question when you get bubbling questions

a. **Water seal- intermittent** bubbling in the water seal is GOOD (DOCUMENT IT)
b. **Water seal- continuous** bubbling in the water seal BAD- there is a leak the system and you need to find it and put tape over it until it stops leaking (LPNs can do this also)
c. **Suction control chamber- intermittent-** BAD- suction is not high enough in that case- need to go to the wall and turn up the dial in the wall so that it become continuous
d. **Suction control chamber – continuous-** GOOD- document it

If something is sealed, should you have a continuous bubbling? NO because its leaking so intermittent is good and suction control is opposite of that

*** A straight catheter (in and out) is to a foley catheter (in, secure it, leave it) as a thoracentesis (in and out to reestablish negative pressure) is to a chest tube (stick it in, secure it, leave it)

RULES FOR CLAMPING TUBES

1) Never clamp a tube for longer than 15 seconds without a dr. order
   -ex: if you break the water seal, you have 15 seconds to cut it off and put tube under sterile water
2) Use rubber tipped double clamps- teeth need to be covered so that tube doesn't get punctured and double because it's best

CONGENITAL HEART DEFECTS

-Trouble or no trouble
-Either it causes a lot of problems or its no big deal at all- no in between at all
-GOOD or BAD

TRouBLe – 7 letters – vowels lower case

Trouble defect- need surgery in order to live
No trouble defect- don't need surgery but might have it years later when it causes any trouble but we wont expect it to have any trouble because its not trouble

Trouble defect-

- Growth and development- DELAYED
- Life expectancy- SHORT
- Apnea monitor because you are in trouble
- In hospital for weeks at a time
- Pediatric cardiologist follows your face

Nurses job is teaching parents the implications not the diagnosis based on whether or not its trouble

-TRouBLe- blood shunts right to left because R comes before L in the word trouble
-TRouBLe-blood shunts left to right in defects with no trouble because that's not how the word trouble is spelt

ex. If a kid has a right to left shunt, what do you tell the parents about surgery?

- right to left blood = TROUBLE

If a kid has a left to right shunt, what do you tell the parents about surgery?

- Left to right blood- NOT trouble

Cyanotic- BLUE (letter B in trouble)- Right to left means BLUE!- TROUBLE
Left to Right means Acyanotic –Not TROUBLE

RECAP
TROUBLE congenital heart defect
-shunts blood right to left
-cyanotic (blue)
-needs surgery
-delayed growth and development
-decreased life span
-needs cardiac pediatrician
-exercise intolerance
-needs meds
-apnea monitor is going to stay longer
-financial guilt
-caregiver stress

NO trouble congenital heart defect
-left to right
-not blue
-no big deal

TRouBle congential heart defect
(T)- first letter – it just so happens that all congenital heart defects that start with the letter T are trouble and if it does not it is not trouble

Examples:
Ventricular Septal defect… trouble or no trouble?
-no trouble because it starts with a V
-shunts blood from the left
-it is acyanotic
-tell parents no big deal

Tetralogy of Fallot… trouble or no trouble?
T- trouble

-Shunts blood right to left
-Cyanotic
-Short life expectancy
-Financial stress
-Growth and development delay

Patent Ductus Arteriosis… trouble or no trouble?
-No trouble
-Shunts blood left to right
-Kid is pink

Patent Foramen Ovale… trouble or no trouble?
-no trouble
-shunts blood left to right
-kid is pink

Truncus Arteriosus
-TROUBLE
-shunts blood right to left
-cyanotic
-etc

….Transposition of the great vessels, triscuspid atresia, anything with a T means TROUBLE!!

…Atrial septal defect, pulmonic stenosis-**no** trouble

ONE EXCEPTION-**left ventricular hypoplastic syndrome- won't be brought up on nclex because its so rare**

ALL congenital heart defect kids will all have these things whether its trouble or not:
1) a murmur because the shunt of the blood
2) echocardiogram done to find out why

4 defects of tetralogy of Fallot mnemonic
Varried pictures of a ranch
VerrieD PictureS O A RancH

VerrieD - **Ventricular Defect**

PictureS - **Pulmonary Stenosis**

Of A - **Overiding Aorta**

RancH - **Right Hypertrophy**

Ex. Your patient has tetralogy of fallot, select all the defects that apply
VD, PS, OA, RH

ANOTHER MNEMONIC: **V**alentines **D**ay **P**ick **S**omeone **O**ut **A** **R**ed **H**eart

Infectious Diseases and Transmission Based Precautions

Standard, universal, contact, droplet, airborne

Contact- anything **enteric**- can be caught from intestine, fecal/oral
CDIff, hep A (Anus), cholera, disenteri, **staph infections**, **RSV** (transmitted droplet but classified under contact precautions because little kids catch it by touching other things that little kids put in their mouth), herpes infections (shingles) (respiratory syncytial virus- fatal to little kids)

HEP **B = (**blood**)
HEP A = (anus**)

Contact isolations- private room preferred- YES to private room, cohort- two RSV kids can be put in the same room (must be **cultured** and **positive** before putting them in the same room, NO mask, gown yes, gloves yes, 425 handwashing, no eyeshields needed unless universal, NO special filter past, NO to patient wearing mask, YES to disposable supplies (plastic utensils, etc.) and dedicated equipment (stethoscope, toys, blood pressure cuff), NO negative airflow

Droplet- for bugs that travel three feet on large particles- all meningitis and H flu-homofluous influenza B can cause epiglottitis

Droplet Isolations- Private room preferred, YES to private room unless you are cohorting based on culture, if they have meningitis they all need lumbar puncture because that's where you culture the meningitis, yes mask, yes gloves, no gown needed, handwashing yes, special eye face shields, no filter mask, pt need to wear mask when leaving room, disposable supplies and dedicated equipment, yes, No to negative airflow

negative pressure room

Airborne- measles, mumps, rubella, tuberculosis, and varicella chicken pox

Airborne isolations: private room REQUIRED unless cohorting, mask yes, gloves yes, gown more for contact, hand washing, no eye face shields, filter mask only for TB, patient leave mask when leaving the room YES, disposable supplied and dedicated equipment not necessary, negative airflow YES.
****TB is spread by droplet but it is on airborne precautions

no face shield
filter only TB

Protect personal equipment- PPE - take it **off in alphabetical order**

1) gloves
2) goggles
3) gown
4) mask

follow the abc's.

OFF is alphabetical ON is **reverse alphabetical for the G's but mask comes second**

1) gown
2) mask
3) goggles
4) gloves

go ~~make~~ ~~meet~~ meet ~~guff~~ goofy guy

Important things to know for math problems:

Dosage calculations- when dr. orders what is not in the bottle and no conversion involved

- Desired/have * available

IV drip rate- volume * drop factor/ time in minutes

**mini/micro drip= 60 drops per mil
**macro- 10 drops per mil

Pediatric dose using child's weight- 2.2 lbs per kg

**pay close attention to amount per day or amount to be given at 1 time

Lecture 4

Crutches, Canes and walkers

One of the major functions of humans is locomotion: testing frequently for casts, traction, crutches, and walkers

Patient teaching is also important- risk reduction

How do you measure crutches- important so that risk reduction is cut down on nerve damage

How do you measure the length of the crutch- 2-3 finger widths below the anterior axillary fold to a point lateral to and slightly in front of the foot

***** if any question says to measure from the axilla or from any landmark on the foot, they are wrong**

Hand grip- can be adjusted up and down and when the hand grips are properly placed the angle of proper flexion will be about 30 degrees

How do you teach crutch gaits?

1) **2-point-** move a **crutch and the opposite** foot together followed by the other crutch and the other foot together (2,2,2,2,2,….)

2) **3 point-** moving two crutches and the bad leg together (3,1,3,1,3,1..)

3) **4 point** – move everything separately (2 crutches + 2 legs = 4)

4) **Swing through** – for none weight bearing- ex. Amputations- or cant bear weight on a leg) – pretty fast- swing themselves

****Amputation with a prosthetic device can bear weight**

When do they use these?

****Even for even odd for odd****

Use the even number gaits (2&4) when weakness is evenly distributed
Use 2 point for a mild problem (mild bilateral weaknesses and 4 point for severe bilateral weaknesses) HOW many legs are affected (2) then pick 2 or 4

Use the odd number gait (3) when one leg is odd

Examples:

1) early stages of rheumatoid arthritis (2 point-systemic disease so both legs should be assumed- early)
2) Left above the knee amputation (swing through)
3) First day post op right knee replacement partial weight baring allowed (3 point)
4) Advanced stages of advanced ALS (4 point- advanced)
5) Left hip replacement second day post op non weight bearing (swing through)
6) Bilateral total knee replacement first day post op weight baring allowed (4 point)
7) Bilateral total knee replacement 3 weeks post op (2 point)

Going up and down stairs with crutches
Up with good, down with the bad

- go up with your good, lead with your good, crutches always move with the bad leg,
- down with the bad, crutches always move with the bad leg

Canes
- Hold the cane on the good side
- Advance it with the bad leg
- When you put the crutch down you have a nice wide stance for support

Walkers

-Pick them up, set them down, walk to them- its slow but this is the right way

-If they must tie their belongings to the walkers, have them tie to the sides, not the front

-No wheels or tennis balls on walkers

Delusions, Hallucinations and Illusions

Nonpsychotic vs. Psychosis

-Very first thing you need to do is decide whether pt is nonpsychotic or psychotic

-Determines treatments, goals, length of stay, medications

Nonpsychotic

-person has insight and is reality based

-emotionally ill

-know they have a problem

-know how its messing up their life

-mentally distressed but not psychotic

Techniques/approaches:

-Good therapeutic communication- right answer for anybody

-Nothing special that you are supposed to know (common sense things)

Example:

Pt Alice says she's depressed and says to you, "I hate this depression and its ruining my life because I have no energy to do anything"- nonpsychotic

Answer: "well how are you feeling now, what is currently making you stressed"

Psychotics-

No insight and is not reality based- don't believe their sick and blame everyone else

-no insight

-treated differently because good communication does not work for them

-unique specific strategies needed

Symptoms

-delusions, hallucinations and illusions are only **psychotic**

-non psychotics do not have any

-delusion- crossed the line and are not in the camp of psychotic

Psychotic Symptoms

Delusion- false fixed idea or belief, no sensory component

1) Paranoid delusion- false fixed belief that people are out to harm you (police, mafia, wife, kids, neighbors, etc.)

2) Grandiose delusion- false belief that you are superior (think your Christ, Mohammed, Genghis Kahn, worlds smartest or greatest person)
3) Somatic delusion- false fixed belief about a body part – I have x ray vision, I can melt stones with my eyes, there are worms inside my arms, you believe your pregnant as an 83 year old male.

Hallucination- False fixed sensory idea – hear, taste, smell, see, touch

1)**auditory**- hear things- voices telling you to hurt yourself (most common reported)
2) **visual**- seeing things that are not there
3) **tactile** –feeling things that are not there
4) **gustatory**-tasting things that are not there (rare)
5) **olfactory**- smelling things that are not there (rare)

Illusion- misinterpretation of reality
-Misinterpreting what's going on through a sensory experience
-Differentiation between illusions and hallucinations- with an illusion there is a referent in realty- something to which a person refers when they same something
-Actually something there but they misinterpret what's there but with hallucination there is nothing there

Examples:

A pt says "I hear demon voices"- example of hallucination because it is sensory and nothing there

A pt overhears a dr. and nurse laughing at the nursing station- and says "I hear demon voices"- example of illusion because they misinterpreted the real sound

During an interview a client says " look I see a bomb" example of hallucination because there is nothing actually there

During an interview a client is looking at the fire extinguisher and says "look I see a bomb" this Is an example of an illusion because they are misinterpreting for a bomb

How do you deal with psychotic symptoms in psychotic patient-
Ask yourself what kind of psychosis do they have?
1) A functional psychosis- they can function in every day life ex. Have a family, children and a job and live alone and take care of themselves but they are psychotic
Example of diseases: schizophrenics & major manics, Schizoaffective disorder, Major Depression (psychotic while depression is not), Manic (acutely)- bipolars are functional but they are not always psychotic

2) Psychosis of dementia – actual damage to the brain and brain is actually damage (in the functional they just haven't learned adaptive behaviors well) but in this care there is actual brain damage ex. Alzheimers, post stroke, organic brain syndrome, senile or dementia falls in this category

3) Psychotic delirium- Functional psychotic- this person does not have brain damage- so they have the potential to learn reality because they don't have any damage, might need medication to balance some chemicals and set structure but they can improve

Role as a nurse: teach reality (4 step process)

1) acknowledge feeling
2) present reality
3) set a limit
4) enforce the limit

ex. What's the first thing you'll say to a patient whom your going to acknowledge feeling- word feeling is used or specification of a feeling
"you seem upset, that's so sad, tell me more about how you're feeling right now"

Presenting reality "I know that ____ is real to you but I do not see it", "I understand those voices are real to you but I do not hear it" or tell them what is real "I am a nurse, this is a hospital and here is your breakfast" either one is good- second thing you do

Setting a limit- "that topic is off limits in our conversation, stop talking about those aliens, we're not going to talk about those voices"

Enforcing the limit "I see you're too ill to stay reality based so I am ending this conversation" stay away from answers like punishment- "since you cant follow the rules you lose your phone privilege" only enforcement is ending the conversation and wont stay reality based

Ex. Schizophrenic (functional) patient says to you, "I'm going to kill you all by morning and I'm starting with you"

1) "I see you are upset" acknowledge feeling
2) "We're going to be kept safe while we're here" present reality
3) "We're not going to talk about that kind of stuff" set limits
4) " I see you are too ill for reality based conversations, so we are going to end this conversation, but we have medication to help your symptom"

However if they have psychosis of dementia- can't learn reality (2 steps)

1) Acknowledge their feeling
2) Redirect them – channel them from something they cannot do to something that they can do – DO
3) Do not present reality because they can't learn it and it'll just frustrate them
4) DO NOT set limits- unfair
5) Problem they usually have: where they are, where their room is, what day it is (NOT PSYCHOSIS , JUST FORGETTING)
6) What technique is not appropriate: do not present reality but DO NOT confuse with reality orientation (tell them person, place and time!!)

EX. Patient with Alzheimer (dementia category)- waiting room of a nursing home on a Sunday, and you say "Mrs. Smith you're all dressed up and she says, "yes my husband is coming to pick me up so that we can go to church" – PROBLEM: husband has been dead for 10 years so she is a FALSED FIXED BELIEF=DELUSIONAL

1) Acknowledge feeling "that sounds like an exciting thing to do" (recognize the feeling if it fits, exploring feeling means asking to try to find out what it is)
2) Redirect "why don't we sit down here and talk about what's going to happen at church today" … ask questions – reinforcement of intact memory –ask to see pictures of grandkids in her room to get her to go back to her room

WRONG ANSWER: that sounds exciting but your husband is dead (only appropriate for schizo's or major manics)

7) Structural brain problem and cannot learn reality
8) Functional can learn reality

Psychotic Delirium- temporary sudden dramatic secondary loss of reality usually due to some chemical imbalance in the body- different than functional- sudden- different than dementia- temporary & secondary

1) People that are crazy for the short term because of something that is causing it ex taking a drug, or high on uppers, or withdrawing from downers, delirium tremors, cocaine overdose, methamphetamine overdose
2) Post op psychosis- withdrawing from a downer, everything goes up, particularly in the elderly- wacky for about 48 hours, looney
3) ICU psychosis- sensory deprivation
4) Cult hidden UTI in the elderly
5) Thyroid storm
6) Adrenal crisis
7) Roid rage (sometimes)

*** TEMPORARY****

Focus: 2 STEPS

1) Acknowledging feeling
2) Reassure that its temporary and they will be kept safe

Removing the underlying cause and keeping them safe

DO NOT present reality because they are not going to get it

EXAMPLE

Functional:

Patient with **schizoaffective** disorder who points to two people talking across the room, and they says, "those people are plotting to kill me"-

1) Say "I see you are frightened" … acknowledge feeling
2) "those people are not plotting to kill you, we're all safe" … present reality
3) "furthermore we are not going to discuss this" … set limits
4) "I see you are to ill to have a reality based conversation so I'll be back in a half hour to try again later" … enforce limits

Dementia:

Patient with **alzeimers** disease who points to two people talking across the room and says, "those people are plotting to kill me"

1) "you seem scared" acknowledge feeling
2) "Let's go somewhere you can feel safer" redirecting

Delirium:

Patient with **delirum tremons** says to you the two people are plotting to kill me
1) "I see youre scared" acknowledge feeling
2) "You are safe and that feeling will go away when you get better" reassure they are safe and will get better

Personality disorders are not considered psychosis they are baseline factors that come along – use good therapeutic communication skills- not classically psychotic

3 clusters for personality disorders:
ABN –Abnormal
Antisocial, **B**orderlines and **N**arcissistic, real sick personality disorders
Treat them more like a functional but use more good communication skills- however functional allows you to set LIMITS

Psych Axis
Axis 1- primary psych disease diagnosis
Axis 2- mental retardation and personality disorders
Axis 3- medical conditions not psych
Axis 4- psychosocial factors like unemployed, recently divorced, newly married, new baby
Axis 5- score- estimation of how high your function ability is

RECAP
Psych questions to ask yourself: psychotic or nonpsychotic?
Nonpsychotic- good therapeutic communication skills
Psychotic- divide into three 1) acknowledge feeling (always first) 2) reality, redirect, or reassure

3 more psychotic symptoms

Your thoughts are all over the place (loosely associated)

1) Flight of ideas- going from thought to though to thought
   Phrases are coherent but they are not tightly connected
2) Word salad- sicker- babble random words
3) Neologism- making up imaginary words
4) Narrowed self concept – when a functional PSYCHOTIC refuses to leave their room or change their clothes because its how they define who they are- WHERE they are and what they are WEARING- don't know who they are unless they are wearing those things in that room- do not force because they will have a panic escalation, instead tell them 1) I see you are uncomfortable or upset, you don not have to leave the room or change your clothes until you are ready
5) Ideas of reference- think everyone is talking about you

**For non-psychotic – ex. Depression, use good therapeutic communication skills
"I see you are depressed and feeling down, its time for you to shower come with me and we will do it" just like you would a post op who just wants to lay in bed

**Only time you're allowed to make choices for patients is for depressed psychomotor patients

Lecture 5

DIABETES

Error of glucose metabolism-cannot metabolize glucose for whatever reason

Glucose- primary fuel source and without that cells die=bad

***Diabetes Insipidus- totally different. Polyuria, polydipsia, leading to dehydration due to low ADH which looks a lot like diabetes mellitus which is why they share the same first name

Best way to remember- like diabetes mellitus only just with the fluids- due to a low ADH- ask do you have a low urine output or low urine output? Both have high urine output

Opposite syndrome of diabetes Insipidus= SIADH=syndrome of inappropriate ADH

Diabetes mellitus has polyuria and polydipsia

SIADH is the opposite so pt would have oliguria and not be thirsty because they are retaining water (gain weight suddenly)

Urine output of 200 ml per hr for 3 hours and a normal blood glucose? Diabetes Insipidus

Urine output of 200 ml per hour for 3 hours and a blood glucose of 280? Diabetes mellitus

10 cc of urine out in 3 hours and a normal blood glucose? SIADH

Insulin lowers the blood glucose

What is the relationship between amount of urine and specific gravity?

- **inverse**
- the less the urine out the higher the specific gravity
- urine value goes up the specific gravity is low

Which would have fluid volume deficit?

- low fluid in the body and high output= DM & DI

Who would have fluid volume excess?

-SIADH

Diabetes type 1 & 2

Type 1
Insulin dependent
K-ketosis prone

juvenile

Type 2-
-Not insulin dependent

adult

-Not ketosis prone

Signs and Symptoms
Polyuria- high urine output
Polydipsia thirsty
Polyphagia- increase swallowing- eating a lot – increased bleeding after a tonsillectomy – hem

Treatment
1) Type 1 without treatment can DIE
D-diet (least important, count carbs, do checks and give insulin accordingly, just lay off refined sugars)
I-Insulin (most important)
E- exercise

2) Type 2 without treatment end up DOA
D-diet (most important, some dr. like for it to be controlled with diet alone)
O-oral hypoglycemic
A-activity

DIET INSULN AND EXERCISE
A) calorie restriction (type two- restrict calorie)
b) Need 6 small feedings a day (split 1800 calories into 6 meals to keep glucose levels and avoid peaks- blood glucose will stay more normal

Example. Type 2 diabetic best diet to follow
a) Restrict calories
b) Divide food into 6 feedings a day

Answer: restrict calories is most important

Best: narrow it down to 2 and think it through- "I will do this one and not do that one and flip it around"- pick the answer you like better

Insulin

***Insulin lowers the blood glucose**

4 types of Insulin you need to know:

1) Regular
2) NPH
3) Lispro
4) Humulog
5) Lantis

Regular (stands for RAPID and RUN- Fast acting and ran in IV)

-onset is in 1 hour
-peak is in 2 hours
-duration is 4 hours
-clear (solution) so it can be IV dripped
-Intermediate acting insulin- because 5-10 years ago we didn't have Lispro and Regular was the fastest at the time
-Still considered as a rapid short acting Insulin

NPH (Not so fast (intermediate), and not in the bag) insulin
-true intermediate acting insulin
-onset is 6 hours
-peak is 8-10 hours
-duration is 12 hours
-cloudy (suspension)- precipitates, particles falls to the bottom over time so it cannot be given IV drip or you will overdose patient and they will die

Example of question for peak:
You gave 30 units of N at 7 am, when would you check for hypoglycemia (when med is at its peak)
Answer: N-3-5pm

Humolog (Lispro) fastest acting insulin
-Onset is 15 minutes
-Peaks at 30 minutes
-Duration is 3 hours
-Given as they begin to eat (**with** meals)

LANTIS (glargine)-long acting
-long acting insulin
-so slowly absorbed that it has no essential peak
-little to no risk for hypoglycemia
-only insulin you can safely give at bed time
-will not go hypoglycemic at bed time so can be given routinely
-duration is 12-24 hours

**check expirations on Insulins- only good as long as its still closed
-Once opened, the manufacturers date is irrelevant, the new expiration date will be 30 days after that (need to right EXP and then the date
-Refrigeration is optional- don't have to refrigerate in the institution
-In the hospital the ones that should be refrigerated are the unopen vials however once a nurse opens two things happen 1) needs new exp date 2) does not need to be refrigerated

BEST answer is-expiration date

**exercise potentiates insulin (does the same thing as insulin)

-think of insulin as another shot of insulin (and he got another shot of insulin—replace with any form of activity in a question)

IF you have more exercise- need less insulin
IF you have less exercise- need more insulin

If a diabetic is going to be active – he better eat rapidly metaboliable carbohydrates

SICK DAYS- flu, diarrhea, etc.
-need to take their insulin even though they're not eating because they are under the stress of insulin
-need to take sips of water because they will get dehydrated
-must stay as active as possible to lower their glucose because even if they don't eat their blood glucose will go up

2 main problems with diabetics
1) Hyperglycemia
2) Dehydration

Complications of Diabetes
Never go to boards not knowing signs and symptoms of the three acute complications of diabetes

1) LOW BLOOD GLUCOSE in a Type 1 or Type 2 – called **insulin shock, insulin reaction or hypoglycemic shock or hypoglycemia-** means glucose is low
    a. What causes this: not enough food**, too much insulin or medication (primary)** and too much exercise-
    b. What is the danger? – Permanent brain damage-vegetative state with one mistake
    c. Signs and symptoms? – **DRUNK** + SHOCK- staggering, slurred speech, poor judgment, slow reaction time, labile(all over the place, laugh cry laugh cry), loud, obnoxious and belligerent, hypoglycemic – cerebral cortical compromise …. **SHOCK**- vasomotor part of the syndrome- low BP, tachycardia, tachypnea, cold, pale, clammy, mottle, patchy
    d. Treatment: administer rapidly metabolizable carbohydrates – SUGARS-ex. Any juice, candy, milk (lactose), honey, icing, jam, jelly. + **ideal combination of food.. sugar+starch or protein** = orange juice + crackers- apple juice + slice of turkey, milk (sugar + protein) but use skim milk because you don't want them burning fats for ketones. HARD to find vein because they are in shock. If they are unconscious give glucagon, dextrose per IV D10 or D50 (D5 wont cut it)
2) **DKA- Diabetic Ketone Acidosis**- only **TYPE 1**(cause another name for Type 1 I ketosis pro and another name for type 2 is ketosis pro)
    a. Causes: too much food, not enough medication, not enough exercise will make glucose go high. But primary cause is **acute viral upper respiratory infections within the last two weeks-** after they recover they start going downhill and getting more lethargic-diabetic ketone acidosis coma

i. Example: if a child comes in with a blood glucose of 250 and type 1- what's the first question you would ask the parents="have they had a viral respiratory infection in the last two weeks? " Because what caused the glucose to get that high was the stress of that illness that was not cut off and they started to burn fats for fuel and got into a negative situation.

b. Signs and symptoms: DKA- **Dehydrated** look (poor skin tugor, warm temperature, hot flushed dry skin), Ketones in their blood (you can have ketones in your urine and not have DKA but is for sure when its in your blood), **kussmaul-** deep and rapid (hyperventilate), high k+ (potassium), **Acidotic** (metabolic), **acetone** breath (fruit odor), **Anorexia** due to nausea because they don't want to eat,

c. Treatments: IV fluids at a fast rate, around 200 an hour, use regular insulin and run it at about 100 an hour, use D5 (only getting 60 calories an hour) will not cause hyperglycemia situation.

3) Low blood glucose is the type 1 is the same as type 2
4) High blood glucose aka HHNK, HHS HHNS, HHNC- TYPE 2 (non ketotic) – any time you see the prefix NON you know it's a type 2- have the nons- type 1 never has anything that has NON in it – this is dehydration!-wherever you see the phrase, hyperosmolar, hyperglycemic nonketotic coma you can pull it out and replace for dehydration, (low water hot flushed dry, Nursing dx. Fluid volume deficit, nursing intervention: giving fluids, goals/outcomes: increased output, moist mucus membranes) DKA without the K or the A-, HHNK is just the D in DKA

Which one is the use of insulin most essential in treatment, HHNK or DKA?-answer: DKA- don't have use insulin with HHNK because they are just dehydrated and you need to just give fluids

Highest mortality rate? HHNK

Higher priority? DKA- because HHNK they come in a lot later because they don't have the ketosis or acidosis that makes people see symptoms and don't see symptoms until they are very bad and get worse--- DKA is acutely ill that can be simply treated wit insulin rehydration while HHNK is not

Who would die first? DKA – lower mortality rate even though its more life threatening as they are treated first

LONG TERM COMPLICATIONS of diabetes are related to

1) Poor tissue perfusion
2) Peripheral neuropathy

Long-term complications

1) renal failure – poor tissue perfusion- losing control of bladder (peripheral neuropathy)
2) ganggreen

3) stasis ulcers
4) blindness
5) heart disease
6) brain disease

Which lab test is the best indicator of long term glucose control?

- Hemoglobin A1C- glycostat hemoglobin- glycosolated hemoglobin-same tests
- Numbers: HA1C- 6 and lower (in control)
- Number that means out of control? 8 and above
- 7? they are on the border so they need a work up and evaluation for some type of infection somewhere- maybe
- A1C- 0.9 change is HUGE

Lecture 6

Drug Toxicities

1) **Lithium**- antimania- bipolar mania- therapeutic level is -0.6-1.2- toxic level is greater than or equal to 2, gray area in between where no books agree on- and is not tested on
2) **Lanoxin** (digoxin)- treats Atrial fibrillation (adena beta calcium dig) and congestive heart failure- 1-2 is therapeutic, toxic Is greater than or equal to 2- value of 2 is toxic because its safer to call something toxic when it may not be-
3) **Aminofilin-** relieves spasms in your airway- (technically not a bronchodilator- doesn't stimulate your beta 2 agonist cell just relaxes a spasm,.. epinephrine is a bronchodilator)- inflammation- theyre in a spasm, acute lock down spasm, need to relieve spasm before giving bronchodilator so that it works better, 10-20 is the therapeutic level, under 10 is not enough, 21=toxic, 20=either way so call it toxic
4) **Dilatin** (phenytoin)- seizure, therapeutic level is 10-20 toxicity is greater than or equal to 20.
5) **Billirubin**- waste product from the break down of the red blood cells, when boards test bilirubin they only test it in newborns not in adult- babies are breaking down moms red blood cells so they are usually high- 8 = no big deal… elevated level= 10-20.. 9.9 and less= normal for newborn but high for an adult, toxicity= greater than or equal to 20- a child with what bilirubin and above needs to come to the hospital? 14-15 doctors start thinking about hospitalization because 15 means you are half way to toxic which will lead to death

**Toxic levels: 2's and 20's
2's -lithium and lanoxin- pick lower number for L's
20's- bilirubin, phenytoin, and aminofilin– go high

Kernicterus- bilirubin in the brain – when your bilirubin crosses your blood brain barrier and crosses your spinal fluid- bilirubin in the brain – usually occurs when you get up around 20- and causes aspetic (no germs) meningitis/encephalitis (baby can die)
Jaundice- bilirubin in the skin
Opisthotonos- position baby assumes when they have bilirubin on the brain- hyperextend due to the irritation of the meninges- babies usually are very flexible but with this, their heels will touch their ears and they are very rigid/extending neck- need to catch it right away-
IN what position do you place an opisthotonos child? On their side

Pathologic jaundice vs physiologic jaundice

Physiologic- bilirubin is normal at birth and kid turns yellow over the next two days
Pathologic jaundice- high at birth and kid yellow at birth

If they come out yellow, something is **wrong (PATHOLOGIC)**
IF they turn yellow over a few days that is **typical (physiologic)**
BLUE BOOK

DUMPING SYNDROME VS. HIATAL HERNIA (opposites)
****Both gastric emptying problems**
Definitions:
Hiatal hernia- regurgitation of acid meaning acid comes back up into your esophagus because the upper part of your stomach herniates upwards through the diaphragm- stomach is two chambered- like a cow- part of it goes up- two chambered stomach with a band around the middle and when you eat, it'll sit in the upper part. Gastric content move in the wrong direction at the correct rate (stomach empties at a normal rate but in the wrong direction)
Traffic violation: going the wrong way on a 1 way street

DUMPING SYNDROME- usually follows gastric surgery in which the gastric content dumped too quickly into the duodenum (gastric content moves in the right direction too quickly)
Traffic violation: going the right way but speeding

Signs and symptoms of hiatal hernia: GERD- Gastro- Esophageal –reflux-Disease (heart burn and indigestion)- hiatal hernia is GERD if you lie down after eating. If you have indigestion after eating it does not mean you have Hiatal hernia-

Ex. Which one has GERD and which one has hiatal hernia?
1) A nurse gets up in the morning, skips breakfast, does meds and treats and then at lunch time gets epigastric burning pain and indigestion and it really hurts – GERD

2) Nurse gets off at 7pm, gets home and eats dinner at 8 and sits down and watches TV and continues snacking and goes to bed and half hour later they have indigestion- HIATAL hernia because they laid down after they ate – dependent upon position and meal times

Signs and Symptoms of dumping syndrome: talk about DRUNK- staggering gait, slurred speech, impaired judgment delay, labile emotions- because you have cerebral impairment because all of your blood is going to the gut because it dumped into the duodenum. Then you also get signs of shock: hypotension, tachycardia, cold, clammy skin, DRUNK + SHOCK=hypoglycemia. To get dumping syndrome, you get acute abdominal distress-signs and signs and symptoms: cramping, pain, guarding, protecting, hear borborygmi, diarrhea, bloating, distention, tenderness, all goes with dumping syndrome, learn them with drunk, shock and acute abdominal distress.

4 of the major things they stress:

Drunk=drunk
Shock=shock
Drunk + shock= hypoglycemia
Drunk+shock+acute abdominal distress= dumping syndrome

To change the way the stomach empties, you can:
-play around with the head of the bed
-play around with the water content of the meal
-play around with the carbohydrate content of the meal

Treatment for hiatal hernia (hint: want stomach to empty faster and why? If its empty it will not reflux) want to higher the head of the bed during and after meals to have gravity empty stomach faster, increase fluid content in meals so that it goes through stomach faster, carbs go through stomach very fast so up carbs
**Hiatal hernia everything needs to be HIGH- head, fluids, carbs

Treatment for dumping syndrome (hint: want stomach to empty slower)
Lower head flat, turned to side and turning to the side with their head down, lower fluids with meals and only give fluids 1 hour 2 before or after meals, lower carb content to slow stomach emptying,
sLOWer: head low, fluid low, carbs low

CURVE question: protein in the diet? Low protein in hiatal hernia, and high protein in dumping syndrome

Electrolytes:

To know your signs and symptoms of electrolytes you need to memorize three sentences:

KALEMIAS: Potassium, do the SAME AS the prefix except for heart rate and urine output
Ex.
Hyper vs. Hypo- HIGH with hyper, LOW with hypo because its doing the same as the prefix EXCEPT WITH HEART **RATE** AND URINE OUTPUT

HYPER signs of the brain: cerebrally UP: agitation, restlessness, irritability, aggression, obnoxiousness, decreased inhibitions
HYPER signs of the LUNGS: tachypnea
HYPER signs of the HEART: slow heart rate, t waves will Peaked/tall, ST's will be elevated
HYPER signs of the BOWEL: diarrhea, borborygmi
HYPER signs of the MUSCLE: spasticity, increased tone
HYPER signs of the Reflexes: +3, +4
ALL SIGNS ARE ALL UP FOR HYPERKALEMIA EXCEPT FOR HEART RATE AND URINE OUTPUT

HYPO signs: lethargy, tachycardia, polyuria, bradypnea, bowels are slowed so you have ileus and constipation, flaccidity in muscles

Questions
Your patient has hyperkalemia, select all that apply:
a) Adynamic ilius
b) Obtundent/stupor (more comatose than lethargy)
c) +1 relfex
d) Clonus
e) U wave (goes down, sign of cardiac depression)
f) Depressed ST
g) Polyuria
h) Bradycardia

CALCIUMS; do the opposite of the prefix

If calcium goes high, everything goes low. If your calcium goes low, everything goes high
HYpercalcemia- bradycardia, bradypnea, flaccid muscles, constipation
Hypocalcemia: tachycardia, tachypnea, agigation, +4 reflexes, seizures,

+2 more things: chvostek and trousseau sign: CH(tap the cheek!) – sign of neuromuscular irritability associated with LOW calcium (opposite of the prefix)
trousseau (blood pressure cuff makes a hand spasm)- French name-an effeminate French men (moves hand like a gay person o.O)

CALCEMIA: think muscles and nerves
POTASSIUM: THINK HEART

MAGNESIUM: Opposite of the prefix-
Hypomagnesemia is associated with hypertension

*Cushing's- need private room because they are immunosuppressed
Hypokalemia –is seen in cushings because you have aldosterone that makes you retain sodium and water – need to kick out potassium for this

In a tie NEVER pick magnesium- if it is skeletal muscle or nerves, blame it on calcium, for everything else blame it on potassium

Examples:

1) Your patient has diarrhea (up), what caused it?
   a. Hyperkalemia (up)
   b. Hypokalemia (down)
   c. Hypocalcemia (up) (not a muscle or nerve problem)
   d. hypomagnesemia (up)- ruled out

   answer: hyperkalemia

2) Your patient has tetany (up), what caused it?
   a. Hyperkalemia (up)-
   b. Hypokalemia (down)
   c. Hypocalcemia (up) – nerve problem
   d. hypomagnesemia (up)- ruled out

Mistakes made with electrolytes:

1) Your patient has tetany, what caused it?
   a. High potassium- same, going the right way
   b. High calcium- going the wrong way, do the opposite (skeletal muscle)
   c. Low magnesium- going the right way, do the opposite

   Answer: high potassium

Sodiums:
Dehydration- HYPERNATREMIA
Overload- HYPONATREMIA

1) A student nurse runs to you and says "I just ran a whole liter of IV fluid in 10 minutes, I forgot to close them clamp." What electrolyte imbalance would you expect to see? HYPONATREMIA
2) ON Lasix? HYPONATREMIA
3) Who is given lots of fluids? HYPERNATREMIA
4) Who has hot flushed dry skin: hypernatremia=dehydration

IN addition to high potassium, what other electrolyte imbalance is possible in DKA?
-hypernatremia because anywhere you see dehydration it will always be hypernatremia

What nursing diagnosis would be major for hyponatremia?

- Fluid volume excess or SIADH
- HHNK- hypernatremia=dehydration
- DI=hyponatremia

**Earliest sign of any electrolyte imbalance is numbness and tingling (paresthesia)
Circumoral paresthesia-numbing and tingling lips

**All electrolyte imbalances cause muscle weakness (paresis)

Recap

Kalemias- same as prefix exept for heart rate and urine output
Calcemias and Magnesemias-opposite as prefixes
E=dehydration-hypernatremia
O-hyponatrami

TREATMENTS:
Potassium
1) NEVER PUSH K+ IV
2) NOT MORE THAN 40 of K per liter of IV fluid (question and clarify)
3) High potassium-stops heart=most dangerous of all- GIVE D5W with regular insulin to drive potassium in the cell and out of the blood- gets K+ down fast- after K is lowered- potassium will begin leaking back to the blood because its just a temporary fix but FAST fix
4) Kayexalete- oral or enema- full of sodium and as it sits in the gut it liberates sodium and makes potassium leave the blood- end up with hypernatremia=dehydration and fixed that problem with fluids – good side: last longer because it gets rid of the excess potassium to never reoccur permanent- downside: takes a long time and you may not live that long

****Give 3&4 both at the same time****

What electrolyte does Kayexalate work with? Potassium
Does it make potassium enter the cell or exit the body? exit the body slowly and late

D5W and regular insulin- opposite- enters and does it really fast

Practice questions- drawing arrows

Lecture 7

Endocrine Glands

Only about thyroids and adrenals

Hyperthyroidsm- metabolism because that's what the thyroid does- regulates metabolism

Logical set of signs and symptoms with high metabolic weight? Weight loss, skinny, high pulse, high BP, irritable, obnoxious, heat intolerance, and cold tolerance, exophthalmos (bulging eyes)

Graves disease=hyperthyroid= you're going to **run** yourself into the **grave**

Treatment: radioactive iodine

1) Pt should be alone for 24 hours
2) Need to be careful with urine (flush three times, hazmat team to clean up spilled urine) because the radioactivity is excreted here …urine is hazardous to nurses
3) No family visitation for 24 hours, home restriction

More treatments:
1) PTU: propylthiouracil
-hyperthyroid- lowers PUTS THYROID UNDER
-primary use is for cancer but also hyperthyroids
-immunosupression- need to watch WBC-

2)Thyroidectomy- remove all or part of it- total or sub
-totals: need life long hormonal replacement and at risk for hypocalcemia because its almost impossible to spare the parathyroids when you take the thyroid out so you end up with low parathyroids and low calcium

Signs and symptoms of hypocalcemia –opposite, everything Is going to go up
2 signs- chvostek's and trousseau's

Sub-do not need life long replacement because they should kick in even if you do hormone therapy for a bit
Not much hypocalcemia
At risk for thyroid storm and crisis (4)

1) Super high temps of 105 and over
2) Extremely high blood pressure (stroke category 210/180)
3) Severe tachycardia (180s)
4) Psychotically delirious

MEDICAL EMERGENCY because it can cause permanent brain damage with hypoxia

Treatment: get temp down and oxygen up

How do you get temp down?
Ice packs (first way)

Best way: cooling blanket without the ice but icepacks come first while you order cooling blanket

How do you get O2 up- oxygen per mask at 10 L – going to be difficult because they are psychotically delirious- must come out of it themselves without treatment or else they will die- **self limiting condition** all we can do is spare their brain until they can get themselves out of it. 2 staff for 1 patient. Can last 2-12 hours.

How would Tylenol work?
Badly because Tylenol works in the hypothalamus and at this point it is being severely threatened because its right near the pituitary and what caused the thyroid storm was the TSH from the pituitary. So the pituitary hypothalamic access is part of the problem. Ice is more important- you just sit there and do what you can but good nursing care will save your life when medicine can't.

****oxygen first** ice pack second and then cooling blanket. Always pick stay with your patient

Post op risks: for both sub and total
First 12 hours: sub or total:
1) Top priority is **air way** because if there is edema near the thyroid it is very bad because it is pressing on the larynx/airway
2) Risk for **hemorrhage** as it is an endocrine gland that has a lot of blood vessels

Longer than 12 hours but less than 48

a) Post op longer 12 but less than 48 for total: tetany (larynx can go into a tetanic laryngeal spasm which would close the vocal cords in a laryngeal spasm and cut the air way and then you die) - due to lost calcium
b) Post op longer 12 but less than 48 for subtotal: storm- thyroid storm

Longer than 48 hours?

a) **Infection-**never pick infection in the first 72 hours after anything

Hypothyroidism

Hypometabolism: obese, flat, boring, dull, cold intolerance, heat tolerance (cannot tolerate what you are) low BP and pule, slow test takers + lower grades, mixed edema

Not enough hormone: so treatment would be to give them hormones like synthroid or levothyroxine
Caution: do not sedate because they are already super slow- you would question **Ambien at HS**- sleeping pill before surgery because they don't need it

Questions NPO-nothing-they're not allowed to have oral pills- would be very susceptible to drugs that sedate them- pt could die without thyroid pill+ suppressant effects of anesthesia

Adrenal cortex
Diseases always start with the letter A or C ***initials***
-Cushings
-Conns
-Addisons

Addisions-under secretion of the adrenal cortex
-Hyperpigmented-very tan
-Do not adapt to stress cause their adrenal gland is undersecreting and the stress so when they undergo stress, they will glucose go down and BP will go down and they will go into shock (time bombs waiting to go off because they decompensate easily)

Purpose of stress response: threat to your brain- to perfuse the brain with blood and raise glucose and raise blood pressure-

Treatment: give him steroids (glutocorticoids – all end in –sone, ex. Prednisone)

Addison's are under so you have to **ADD SONes** to them.

Cushings syndrome: oversecretion of the adrenal cortex
-if you have lots of money in your bank around it is CUSHY
-Cushy chair has MORE stuffing

Signs and Symptoms: two things
1) Signs and symptoms of Cushing's
2) +all side effects of steroids (Sones)

Draw picture of a little man- CUSH Man – draw smell head (puffy moon face, beard (Hirtuism), big body (bump on front(Gynecomastea-female breast on men), and bump on the back, trunk or central obesity with skinny arms and legs), skinny arms and skinny legs (atrophy of those muscles because they waste away) fill in full of water (retaining sodium and water)- losing potassium out the back, striae on abdomen then … High glucose because they are going to be hyperglycemic (look like diabetics) , bruises very easily, IM mad I have an Infection in thought bubble, which means he's irritable grouchy and immunosuppressed,

What happens if your on a steroid and diabetic?

- Need more insulin because the steroid is going to increase blood glucose and makes person go crazy
- Accuchecks Q6- because steroids make your insulin go up

Treatment:
What is the treatment for hypersecretion of a gland?

- cut it out- adrenalectomy would be your treatment

why would doctors do this?- if you look like Cush man and if you get Addisons you would be on a steroid that ends in –sone and the side effects would make you look like cushman which is why its so frustrating because they will take a long time to look and feel normal (about a year)
-endocrine surgery creates the opposite problem for which you have to take the hormone the side effects make you look like the thing you had in the first place for example if you're hyperthyroid- treatment is thyroidectomy- you induce hypothyroid-need to take thyroid pills and side effects would look like graves disease

TOYs for kids

3 things to consider
1) is it safe
2) is it age appropriate
3) is it feasible

Safety considerations!

1) No small toys under 4 year
2) No metal (die-cast) toys if oxygen is in use because it can cause sparks
3) Beware of fomites- nonliving object that harbors microorganisms (toys are constantly going in their mouth, toys are the worst fomites, hard plastic is best because you can bleach it till they are dead)
   -Living organism-vector or a host

If you have a child that is immunosuppressed, what would be the best toy?
-hard plastic action figure

feasibility-could you do it? 13 year old can swim and its safe and age appropriate but its not feasible for a 13 year old in a body cast

Age appropriateness
1) **infancy 0-6months**- best toy is a musical mobile because they are sensory motors so its best that it stimulates both motor and sensory
-second best is to look for something soft and large so they don't choke or hurt themselves
2) **6-9 months** – working on object permanence- still there even if you cannot see it- cry because its gone- need to teach them this- cover uncover toy- jack in the box, pop up pals, books with little windows, peek a boo
-second best: something large but firm-
-worst: musical mobile because they will try to reach it and accidentally hang themselves
books: peaked window books
3) **9-12 months**: vocalization –learning to speak – speaking toys are best ex. Woody the cowboy, tickle me Elmo, barnyard friends, talking books, purposeful activity toys (building blocks, stack etc.)
   - **never** pick an answer with the **following words if kid is under 9 months**- build sort stack make construct

4)**1-3 year old- toddlers**- push pull toy- lawn mower, wagon, buggy, baby stroller, dog with flappy feet you drag- working on gross motor skills (running, jumping)
-if it takes finger dexterity do not choose it – cannot use colored pencils or scissors, finger painting means hand painting – characterized by parallel play play alongside but not with
5) **Pre schooler 4-5 yrs-** work on their fine motor skills- things that take finger dexterity- work on their balance- tricycles, tumbling class, dance class, ice skates
characterized by cooperative play- play with each other in groups – like to pretend and highly imaginative, all need the same prize
6) **school age 6-13**: characterized by 3 C's- creative (let them make it- blank paper with colored pencils rather than coloring book, cook, legos a great toy because they are building and making) Collective- always collecting something, Competitive: they like to play games where there is a winner and a loser
7) **adolescents**: peer group association- want to hang out with friends and do nothing – if they gave you a question where a gang of teens were hanging in the room unless 1 of these 3 is true: 1) if anyone is fresh post op (12 hours) – 2) if anyone is immunosuppressed 3) if anyone has a contagious disease

Laminectomy
Lamina- vertebral spinose processes (not the round body, the wing things instead)
Ectomy- removal

Reasons: to relieve nerve root compression- calcium or herniated discs pressing on them- but away bone to give more room for nerve to exit

Signs and symptoms THE 3 P's
1) Pain
2) Parestheisa (numbess and tingling)
3) Paresis (muscle weakness)

Most important thing in any neuro question is location- determines the prognosis, treatment, symptoms

3 locations
1) Cervical –neck
2) Thoracic –upper back
3) Lumbar –lower back

Example:

1) What is the most important thing to check pre-op for a cervical laminectomy
   Interventions
      1) check breathing because if it is cervical, it means that it is intervening the diaphragm
2) The function of arms and hands is back up answer

3) What if the questions is asking about thoracic

   a. Cough mechanism (will not allow you abdominal muscles to contract when you cough) and the bowel is most important
4) What if the question is asking about lumbar
   a. Affects the bladder and the legs-is their bladder distended or empty and back up is the function of their arms and legs

Post op laminectomy/spinal cord-log roll (best answer ever)

- **-do not dangle**- do not let them sit on edge of the bed- go from lying down immediately to walking around-should not sit there for 10 or 15 minutes dangling – **NO SITTING**
- **do not sit for longer than 30 minutes**-typical post op order? Meals only sitting time … they may walk, stand and lie down without restriction
- What worker in the hospital has the highest risk for back injury? Admitting clerks because they sit all day- sitting is very bad for your back

Post op complications- locations
Cervical- wont breath very well or deeply- can develop pneumonia- location closest to lungs
Thoracic- wont cough- can develop pneumonia and can develop and ileus for their bowels not working
Lumbar- urinary retention followed by problems with the legs

These apply to anything Neuro-location very important

Anterior thoracic will have chest tube
-front through the chest you go to the spine – will have a pneumohemothorax

Laminectomy with fusion

- Bone graft from the iliac crest- hip

- Cannot have bone on bone so there is fusion to avoid grinding
- Pt will have two incisions on hip and spine
- -Which incision will hurt more? The hip will
- Bleeding and draining? Hip with drainage
- Infection: both equal
- Higher risk for rejection: spine
- Hip causes the most problem- shorter recovery to get rid of hip incision
- Surgeons are now using cadaver/ synthetic bonding substances bones from Banks- to avoid second incision- slight risk of infection

Discharge teaching: 4 temp restrictions and 3 perm

1) Do not sit for longer than 30 minutes for 6 weeks (usually right)
2) Lie flat and log roll for 6 weeks
3) No driving for 6 weeks
4) Do not lift more than five pounds for 6 weeks (gallon of milk)

1) Cannot lift objects by bending at the waist, always with the knee
2) Cervical lambs not allow to life anything over their head forever- need step stools
3) No off trail biking, amusement park rides etc.

LECTURE 8

Lab Values

High priority lab values

A) Abnormal but not a priority (don't have to do anything about it, you can ignore it and have the doctor discover it in the morning and there would be no troubles
B) Abnormal and need to be concerned but still nothing to do, just watch them
C) High Priority- critical and must do something about it
D) Highest priority that you can possibly have with a lab value

Creatinine- best indicator of kidney or renal function (SERUM) 0.6-1.2 same (numbers as the lithium range) **Level A** – have kidney disease but that's fine, only make a phone call if they had a test the next morning that involved dye

INR-International Normalized Ratio- monitors coumadin therapy, like the pt (variation of the prothrombin time)- normal range is between 2-3, anything 4 and above is a **Level C-** critical and you need to do something- whenever you get a situation on what you need to do for something, there is a protocol you need to follow:

1) Always HOLD- if there is something that's causing a problem stop it,
2) Asses- focused assessment on the area the lab value is telling you there is a problem with

elevated
Blood values
you can always
pick
dehydration

3) Prepare to give- whatever you need- don't always give just prepare
4) Call whoever is appropriate

Example: HOLD Coumadin, assess for bleeding, prepare to give vitamin K and call your doctor

Potassium- indicator that something is wrong- 3.5-5.3-
a low potassium is a **Level C-** don't need to hold anything, assess the heart, prepare to administer potassium, and call the doctor
a high potassium level 5.4-5.9 is a **Level C-**hold off potassium, assess the heart, prepare Kayexalate/D5W/Regular Insulin and call doctor- if the potassium is greater than or equal to 6- it is DEADLY- Do everything you'd do for a level C and call STAT- have a nurse handle every task- cannot leave the bedside of a D but you can for a C- so remember you stay at the bedside and everyone else helps you get it done

pH-7.35.7.45- a pH in the 6's is a **LEVEL D-** Assess vital signs because as the pH goes down so does your patient- doing vitals to make sure their still alive- cannot prepare anything-to correct it is to treat the underlying cause- the physician has to get here to determine the cause- assess vital signs and call the doctor. STAY AT BEDSIDE

BUN-nitrogen waste products in the blood- 8-25- Level A BUNs (8 hotdog buns in a pack)- must assess for dehydration

anemia **Hemoglobin-12-18-** if its 8-11- it's a **Level B –** and you assess for low hemoglobin/bleeding/malnutrition but if it falls below an 8 is a **Level C** and you must do something- assess for bleeding, prepare to administer blood and call the doctor

BICARB- 22-26- abnormal is a **LEVEL A** 2+2+2=6

CARBON DIOXIDE- arterial blood gas analysis- 35-45- high but in the 50's- **LEVEL C- critical (for people without COPD)** assess for respiration status, make pt practice pursed lip breathing because they are exhaling, most of the time it will correct the problem because they will breath easier
High and in the 60's- diagnosis for respiratory failure- CO2 is 60 and above- medical emergency- do not leave the room, **LEVEL D-** assess respiratory status, prepare for intubate and ventilate and then call respiratory therapy first and then call the physician but stay with the patient

Hematocrit- 36-54- 3 times the hemoglobin- elevated hematocrit is abnormal at a **level B** and assess for dehydration

PO2- from the blood gas analysis- 78-100- if it is low but still in the 70's then it is a **Level C-** assess respiratory, give them oxygen and it will likely correct it as the dyspnea/tachycardia and restlessness goes away.

If its low in the 60's it's a **Level D** and in the same path as respiratory failure- 2 defining characteristics- CO2 the 60's and O2 in the 60's need to intubate and ventilate- nothing to hold, assess respiratory status, prepare to intubate and ventilate, call respiratory therapy and you call the doctor- can put oxygen even though it wont solve the problem- throw on the O2- call respiratory therapy and call the doctor

Ex. When someone is hypoxic- which rate increases first? Heart Rate increases first then when the heart can no longer compensate then your respiratory rate will

2 most common causes of episodic tachycardia in heart patients- hypoxia and dehydration- increased IV rate and give them some oxygen and you will probably not have to call the doctor

*Assess before you do unless delaying doing puts your patient at higher risk- if you delayed stopping the blood you would be putting the person at increased risk- assess before you do unless delaying doing in order to assess puts the patient at risk-

Example:
A patient pulls out their arterial line and are bleeding in bright red spurts from their radial Artery, what would you do?
-Assess their vitals second, apply pressure first, do will proceed to assess because your patient is at risk

Acute dyspnea- elevate head of bed first, if you do a respiratory assessment with head flat you're not going to get good data- like to position first- usually position works over the other actions- almost always position first when you're in between to Do's. However in the BEST question: you must give oxygen first. This only applies to FIRST question.

O2 Sat should be 93-100- anything less than 93 is a **Level C-** unless they're in real danger there is no reason to throw on the O2- in peds freak out if its below 95 because little kids don't desaturate- SO2- anemia falsely elevates it- or a dye procedure in the last 48 hours because of the dye colors

BNP- brain natruetic peptide- best indicator of congestive heart failure and should be under 100- over 100 is a **level B- just watch for CHF because it indicates a chronic condition not an acute one**

SODIUM- 135-145- abnormal is B- assess for dehydration if high and low assess for overload/FVE- if the question tells you the sodium is abnormal and there is a change in LOC- the priority of the patient is now a **level C-** dangerous Safety

White blood cell – 5,000-11,000

ANC-absolute nutriphil count- needs to be above-500 per cubic ml – no range but ne

CD4- needs to be above 200- if it falls below that means you have AIDS-

Low white counts- level C- all are and because of that there is nothing to hold and you assess for signs of infection and place them on neutropenic precautions: don't drink water that has been standing for longer than 15 minutes- not allowed to have water pitchers- not allowed to have a water bottle,

Platelets- no normal range- key is trigger values for thrombocytopenic precautions? A platelet count below 90,000 is a level C and you need to put them on bleeding precautions and less than 40,000 Is Level D

RBC- 4-6 million- count is just a Level B

***memorize the 5 D's**

1) pH in the 6's
2) potassium in the 6's
3) CO2- 60's
4) O2- 60's
5) platelet count of less than 40,00

***memorize the 8-10 C's and what to do for them**

1) Carbon dioxide in the 50's – assess respiration status and make patient practice pursed lip breathing and that should solve the problem
2) INR- anything 4 and above - HOLD coumadin, assess for bleeding, prepare to give vitamin K and call your doctor
3) Potassium lower than 3.5- don't need to hold anything, assess heart, prepare to administer potassium, call doctor, High potassium: lower than 6: hold anything with potassium/insulin, assess heart, prepare to administer kayexelate, call doctor
4) PO2 in the blood gas analysis- low but still in the 70's- assess respiratory status and administer oxygen and it will make dyspnea/tachycardia/restlessness go away if it worked
5) Sat O2- anything less than a 93- anemia falsely elevates it and so do dyes
6) Sodium- if question says there is low or high sodium and LOC is affected, either add or take away sodium
7) Low white blood cell counts- follow neutropenic precautions- ex. Cant drink water that has been standing still, need room cleaned daily, etc.
8) Low platelet counts under 90,000- need to put them on bleeding precautions

Lecture 9

Psych drugs

Psychotropic drugs- big laundry list of psych drugs you need to know but they have a lot in common

*ALL psych drugs cause low blood pressure and weight changes (gain mostly)- mostly increase but Prozac and a couple others can make you lose weight

Phenothiazines

-First generation antipsychotics/typical antipsychotics
-All end in zine
-Do not cure psych diseases just reduce the symptoms
-In large doses they are antipsychotics
-"We use zines for the zany"
-In small doses they are antiemetics: reduce nausea but if you double the dose it will treat psychosis
-Considered major tranquilizers – the big guns
Aminoglycosides are to antibiotics like phenothiazines are to tranqulizers … pull these out when nothing else works

SIDE effects:
A) anticholinergic- primarily dry mouth
B) blurred vision- risk for injury
C) constipation
D) drowsiness- risk for injury
E) EPS- extraparamydial syndrome (like parkinsons)- risk for injury
F) photosensitivity
aG) agranulocytosis-low white count (immunosuprsessed)

*When a patient displays a side effect you teach the patient, inform the doctor and keep giving the pill

For EPS- you would give Parkinson drugs to treat that symptom however it will make dry mouth and constipation worse

Toxic Effects: hold the drug and call the doctor immediately

When a client is on a drug that it's a tranquilizer their diagnosis is that they are at major risk for injury

***Decanoate:** after the name of a drug it means that it is long acting- one shot for a month IM form given to noncompliant clients – if client does not go in for psychiatry or to community mental health center to get their shot, they can put out a warrant for him because the patient is violating court order and they will arrest them, give injection and they go home

Trycyclic antidepressents- old class and most have now been grandfathered into the NSSRI- Non selective Reuptake Inhibitors- chemically they are so much like them
-Mood elevators
-(ELEAFIL=ELEVATE mood)
-Side effects: start with the letter E
A) anticholinergic (dry mouth mainly)
B) blurred vision
C) constipation
D) drowsiness
E) Euphoria
-Must take for 2-4 weeks before you get beneficial effects
-Most will think its not working

Benzodiazepines
-anxiety meds
-minor tranquilizers
-Always have ZEP in the name
-both tranquilizers- zeps and z's are tranquilizers- think of ZzZzZ…
-Zeps are the minors- Z's are the majors
-Zepline- rock concert- a bunch of minors on tranquilizers
-indications: more than just tranquilizers
-pre-op to induce anesthesia, good for alcohol withdrawal, muscle relaxer, good for seizures, help people when they're fighting ventilator because it calms them down
-work quickly but cannot take them for more than 2-4 weeks-

whats the relationship between and antidepressant than a minor tranquilizer- one takes 2-4 weeks to work but you can be on them for life, the other works right away but you cannot be on it longer than 2-4 weeks
-If someone is anxiously depressed they'll give both because the tranquilizer will work right away and then when the antidepressant kicks in they'll take them off the tranquilizer
-Heparin is to coumadine as a tranquilizer is to an antidepressant… heparin works right away but you cant be on it right away and coumadine takes a while to kick in but you can be on it forever. Most patients will be on both until coumadine kicks in

Side effects
A) anticholinergic effects
B) blurred vision
C) constipation

D) drowsiness

The nursing dx is risk for injury

Monoamine oxidase inhibitors
-dirt cheap
-drug names: beginnings of the name all rhythm – lar nar par- larplan, nardil, parnate- that is their trade name not the generic
Side effects:
A) anticholinergic
B) blurred vision
C) Constipation
D) drowsiness

Patient teaching
Letter A- to prevent severe acute sometime fatal hypertensive crisis- patient must avoid all foods containing tyramine
-Foods that have tyramine: fruit and veggie: do not have tyramine but 3 do: Salad BAR- banana, avocado, raisins- any dried fruit
grains are fine
meats: no organ meats: kidneys, tripe, intestine, lung, tongue
-No preserved meats: nothing cured, dried or pickled, smoked, or hot dogs or certain processed lunch meats: not allowed to eat: "and other assorted parts"

Dairy: no cheese except for cottage or mozzarella (aged cheeses are no good), yogurt is bad

No alcohol or chocolate- no caffeine soy sauce etc.

Teach pts not to take over the counter meds

LITHIUM
-treats bipolar disorder because it decrease mania not the depression
-the most unique because all other psych drugs are neurotransmitter messerwithers altering neurotransmitters, but lithium does not mess with neurotransmitters instead stabilizes nerve cell membranes
-Side effects: act more like an electrolyte: The three P's
-Peeing, Pooping and Parasthesia (the earliest sign is numbness and tingling for electrolyte same thing for lithium)
--Give and don't call the doctor

Toxic effects
-hold and call dr.
-tremors, metallic taste and severe diarrhea

A) Intervention: increase fluids- because theyre peeing and pooping all the time which means they are losing fluid and are at risk for fluid volume deficit/dehydration so you must watch sodium
B) Sweating do not give them free water- need Gatorade

C) Linked to sodium so you must monitor sodium intake: low sodium makes lithium toxic and high sodium makes lithium not work- needs to be a normal level

Prozac: selective serotonin inhibitor

- Similar to elavil- same set of side effects belong-
- A) anticholinergic
- B) blurred vision
- C) constipation
- D) drowsiness
- E) Eurphoria

Prozac causes insomnia- give it before noon and never at bed-time
-watch for increased suicidal risk in adolescents and young adults
-what if an adolescent is on prozac, are they at risk?
-- No, because you have to have recently changed the dose and they have to be an adolescent or a young adult- need to keep the dose the same

Hadol
-Decanoate form too-long acting IM
-same as thorazine
Side effects:
A)anticholinergic
B) blurred vision
C) constipation
D) drowsiness
F) photosensitivity
aG) agranulocytosis

*MUST KNOW NMS: elderly patients and young white schizophrenics may develop NMS from an overdose

Neuroleptic malignant syndrome: potentially fatal hyperpyrexia (fever) with temps of 106-108- but it can start at 102- dose for elderly should be a half dose

-Anxiety and tremors (EPS: side effect no big deal) (NMS: fatal big deal that the only difference is taking a temperature: squad needs to come but not much nurses can do just don't leave bedside)

Safety concerns related to the side effects

1) they're running around and then suddenly fall

CLOZARPINE (CLARZAPIL)

- Original first
- Used to replace the zines in Haldol

- Does not have the side effects a b c d e or f – but can
- Disadvantage: agranulocytosis; trashes your bone marrow and get lots of infections: alter formula like to take this away but its difficult because this does not happen to all patients: A typical antipsychotics white count is a big deal

Geodon: black box warning
-Prolongs the qt interval and can cause sudden cardiac arrest- should not be used with those with heart problems
-tranquilizers: zine – new major minor zep zapine and zine

Zolaft- sertraline: causes insomnia but can be given at bed time- interaction
-interferes with cp450 system and increases toxicities and other drugs cannot get metabolized; when you add sertraline you need to lower the dose
-St. Johns warts- will get serotonin syndrome
-SADHEAD – sweating, apprehension (impending sense of doom), dizziness and headache-
-interaction with warfarin- for increased bleeding because coumadine is going to be toxic.

Lecture 10

Maternal Newborn

Pregnancy – Due date

- -Take the first day of the last menstrual period add 7 days, subtract 3 months
- example: June 10-15- March 17 is the due date
- -Weight gain- need to know how much weight a woman should have to gain- 28 lbs + or – 3
- -First trimester- 1 pound each month- 3 months long

- -Second and third- 1 lb per week- need to be able to predict what her weight gain should be
  - Example: 28th week – she has gained 22 lbs, what is your impression?
- --Week 12 is end of first trimester- 3 lbs
- -13 weeks- 4lbs, 14 weeks- 5lbs, 15 weeks- 6 lbs, 16 weeks- 7lbs, 17 weeks- 8 lbs, 18 week- 9 lbs +10= 19 lbs
- -Difference of 9 in each so all you have to do is 22-9=19lbs- she gained 3 more than she was supposed to so she should be assessed- 3 lbs off- just assess but if it was 4 lbs off that means something is wrong
- -ex. 31 week and gained 15 lbs- 31-9=22, so there is a difference of 7 lbs so she would need a biophysical profile on the baby because the baby may have died last month
- -Ideal weight gain=3-9

- **Fundal height**- top part of the uterus- not palpable until week 12, cannot palpate during the first trimester
- When is the fundus at the belly button=20-22 weeks of gestation
- That's important for a nurse to know- because your dealing with date of viability
- Can you use fundal height to determine what trimester woman is in- good for when there is an emergency
- Mom is priority if she's in the first trimester
- If your able to palpate the fundus that means she's in the second trimester and she is still the priority
- If the fundus is above the umbilicus- she is in the third trimester- and the baby becomes the priority

Signs of Pregnancy

- **Positive** and everything else
- Fetal skeleton on x-ray
- Fetal presence on ultrasound
- Auscultation of a fetal heart rate of 140- it's a baby- somewhere between 8 and 12 weeks it can actually be heard but it starts at week 5
- "When would you first", "when would you most likely", "when should you by"- be careful reading OB questions
  - ex. When would you **first** auscultate a fetal heart- 8 weeks- beginning of range
- When would you **most likely** auscultate a fetal heart- 10 weeks- pick middle range
- When **should you** auscultate a fetal heart- 12 weeks- end of range
- When is quickening- when the baby kicks- 16-20 weeks, most likely- 18weeks, when should you- 20 weeks
- When the examiner palpates fetal movement- not when mom does

Maybe signs

- All urine and blood test are maybes (a positive pregnancy test is not a positive sign of pregnancy- its only probable- it only means you have the hormones increase with pregnancy
- Chadwick's, goodell's, hegar signs- occur in alphabetical order – weeks vary from woman to woman but the order does not vary
- -chadwicks- cervical color change to cyanosis- all start with the letter C
- goodell's- cervical softening
- hegar's- uterine softening

Patient teaching

- Teach woman the pattern of office visits to prevent mortality
- Once a month until week 28-
- Week 28- once every 2 weeks until week 36
- Week 36- every week until delivery or until 42 weeks where you would schedule C sections
- Hemoglobin will fall and that's normal- 12-16 is normal, but it can fall to 11 and be perfectly normal – not low but tolerable
- -2nd trimester it can drop to 10.5 and be normal and in 3rd it can drop to 10 and be normal
- Discomforts: morning sickness first trimester- dry carbohydrates before you get out of bed, how do you deal with urinary incontinence- first trimester and third trimester- void every 2 hours all the way throughout her pregnancy until 6 weeks after delivery
- Difficulty breathing- 2nd and 3rd trimester problem- teach tripod position- feet flat, arms on the table leaning forward
- Back pain- 2nd and 3rd trimester- worse and worse- pelvic tilt exercises – tilt the pelvis forwards- have them put foot on stool to get the pelvis to tilt forward
- Good health patterns and ideas

Labor and Birth

- Onset of regular and progressive contractions
- Dilation-opening of the cervix 0-10 cm – 10 cm is 4 inches
- Effacement- thinning of the cervix and it goes from thick to 100 percent
- Woman begins labor thick and closed or 0 cm
- She ends labor fully dilated and fully effaced- 10 cm and 100 %
- Station- relationship of the fetal presenting part to moms ischial spine-smallest diameter through which the baby has to fit to be born vaginally-
- Negative stations-presenting part is above this tight squeeze
- Positive stations- presenting part is below this tight squeeze- good

- if baby stays at negative 1 or 2- for 17 hours after fully dilated or effaced-
- +3 +4= baby can be born vaginally because its already made it through and just needs help with forceps or episiotomy
- Positive numbers are positive news, negative is negative news
- Engagement is station 0- presenting part is at the ischial part
- Lie- relationship between spine of mom and spine of baby- if mom spine is straight and so is baby, that is a vertical lie
- Transverse- bad baby spine perpendicular to moms spine- trouble!
- Moms spine and baby's spine are parallel that is good
- Presentation- part of the baby that enters the birth canal first
- Most common presentation is ROA- or LOA-pick because those are most common- pick R before L

4 stages of labor and deliver

- Stage 1- all of labor and labor has three phases: latent, active and transition
- Latent- LAT- initials of phases in order
- Stage 2- delivery of the baby
- Stage 3- delivery of the placenta
- Stage 4- the recovery – last for 2 hours

What is the purpose if uterine contractions in the first stage? Dilate and efface the cervix

What is the purpose of uterine contractions in the second stage? Push the baby out

What's the purpose of uterine contractions in the third stage? Push the placenta out

What's the purpose of uterine contractions in the fourth stage? Stop bleeding

When does post partum technically begin? 2 hours after delivery of the placenta

Number 1 priority in the 2nd phase of labor- pain management

What's the number 1 priority in the 2nd stage- clearing baby airway

What nurses action in 3rd phase: checking dilation, helping with pain

What are nurses action in 3rd stage: checking for blood loss and placenta parts
D- first stage is called labor!

Labor chart:

- **Phases:** 3- latent active and transition

- **Latent-**0-4 cm, contractions every 5-30 minutes apart, contraction duration is 15-30 seconds and intensity is described as mild
- **Active**-5-7 cm, contractions is 3-5 minutes apart, contraction duration is 30-45 seconds and the intensitiy is moderate
- **Transition-**8-10 cm- contractions are 2-3 minutes apart, duration is 60-90 seconds and the intensity is strong

Example:

- Woman comes in and says 5cm and they last for 45 seconds- what phase is she in? Active

***Only memorize active labor to memorize them all- anything less intense would be latent- 5,7, 3,5 30 60

Contractions should not be longer than 90 seconds or closer than every 2 minutes- this means big TROUBLE

What are the signs of uterine tetany? contractions longer than 90 seconds and closer than every 2 min STOP PITOCIN

Frequency- beginning of one contraction's to the beginning of the next- beginning to beginning

Duration- beginning to end of one contraction- A to B or C to D

Intensity- strength of contractions and its purely subjections

***Teach to palpate with one hand over the fundus with the pads of the finger

COMPLICATIONS OF LABOR
18 different kinds- only 3 protocols you need to memorize

1) Painful back labor- LOP-OH PAIN- position then push, place her in the knee to chest position because it brings the baby down off the sacrum and coxyc (LOW PRIOTY)
    a. push into her sacrum with your fist (2)
    b. position knee chest- ass up face down (1)
2) Prolapse cord- medical emergency- when cord is the presenting part because baby is pressing on it (HIGH PRIORITY)
    a. push the head off the cord
    b. position mom in knee chest position
    c. stay that will until they pull it out csection
3) Interventions for all other complication in labor and birth
    a. LION-

b. L- turn on left side
c. I- increase IV
d. O- oxygenate them
e. N- Notify physician

PIT- in a crisis- if Pitocin is running, stop it
Stop the pit and then LION

Pain medications and labor- do not administer a pain med to a woman in labor if the baby Is likely to be born when the med peaks – (peak 15-30 minutes after you give it)

Ex. you have a prima gravida at 5 cm who wants her IV push pain med- then the answer is yes because the first time mom at 5 is not going to give birth any time soon. Different for a multi gravida at 8 cm because she could deliver in the next hour

Lecture 11-

Fetal Monitoring Patters (7)

1) Low fetal heart rate – under 110- when you see it you do LION, stop pit if running
2) High fetal heart rate- over 160 – document and take moms temperature because mom can have a fever
3) Low baseline variability- bad- fetal heart rate stays the same and does not change- stays in the middle- LION
4) High baseline variability- heart rate is constantly changing-if your vital change that's good before you're born- don't like to see stabile vital signs
5) Late decelerations- near the end of a contraction- heart rate slows down near the end of the contraction – LION

6) Early decelerations- babies decelerates before contraction or at the beginning of the contraction- normal, just document it
7) Variable decelerations- VERY BAD- prolapsed cord- push, position

*Variable is the most unique, VERY BAD- treat with push and then position
3 are good
3 are bad- begin with the letter L and LION begins with L

What are they and what they mean: VEAL CHOP
V ariable- cord compression **C**
E arily decelerations- head compression **H**
A cceleration-Ok- **O**
L ate deceleration- Placental insufficiency **P**

****Ace of Spades- answers that win all the time:
Check fetal heart rate- no matter what happens in OB always check fetal heart rate

DELIVERY OF THE BABY- 2nd stage
1) Deliver the head-cephallic
2) Suction mouth and the nose (alphabetical)
3) Check for a nuchal cord- around the neck
4) Deliver the shoulders and the body
5) Baby must have an ID band on before it leaves the delivery area

Delivery of the Placenta- 3rd stage
1) make sure its all there
2) check for a three vessel cord, how many of each – 2 arteries and one vein- AVA- woman's name

Recovery- 4th stage first two hours after delivery of placenta
4 things you 4 times an hour in the 4th stage
1) Vital signs assessing for signs and symptoms of shock- pressures go down, rates go up and you look cold and clammy- shock
2) Check fundus- if its boggy, massage it and if its displaced you must cafeterize it
3) Pads-check perineal pads to check bleeding- if excessive it will saturate in 15 min or less- it has to be 100 % saturated in order to be in big trouble
4) Role her over- check for bleeding underneath of her- because you can saturate half a pad and bleed out without showing- do these every 15 minutes in this stage

POST PARTUM Assessment – every 4 – 8 hours
1) bubble head
    a. breast
    b. uterine fundus (firm and midline, boggy must massage)
        i. fundal height=day post partum-

1. **example: 4th post partum day, where will the fundus be, 4 below on the fourth day but still midline**

c. Bladder
d. Bowel
e. **Lochia**- rubra, serosa and alba- 4-6 inches on pad is OK- second most important
f. Epesiotomy- incision
g. Hemoglobin and hematocrit
h. **Extremity check**- third most important- looking for thromobophlebitis- bilateral calf circumference measurements
i. Affect- emotional
j. Discomfort

Variations in the newborn-NORMAL

- Erythema- rash-
- Caput succedaneum- CS- crosses sutures and is symmetrical
- Cephalus hematoma-CH- does not cross sutures
- hyperbilirubinemia- normal physiologic appearance after 24 hours

OB meds-don't need to know much-

Tocolytics- stop labor-(causes maternal tachycardia) and mag sulfate(stops labor, hypermagenesemia-high magnesium makes everything go down, heart rate will go down, bp will go down, **reflexes** go down, **respiratory** rate will go down, LOC will go down- your parameters for titrating the mag sulfate- as long as resp are above 12 it is good- want +2 reflexes, if its +1 slow it down and if its +3 speed it up, DO not want to see 11 & + 1) for women who are threatening prematurity

Oxytoxics- stimulate and strengthen labor- Pitocin- can cause uterine hyperstimulation- which would be defined as longer than 90 seconds and closer than every 2 minutes
Methergine- causes high blood pressure

Fetal lung maturing meds- betamethasone- steroid- given to the mother and given IM, it is given before the baby is born, can repeat it as long as baby is in utero
Cirfactant- given to the neonate, not to the mother, given transtracheal, blown in through the trachea, given after baby is born, not before

Medication helpful hints

What is humulin 7030? mix of insulins of Regular and N

70 and 30 are percentages- which one is which- 30 percent is R- 100 units of 7030 how many units of N will there be?- 70 and R will be 30. If it was 50 units, it would be 35 units of N and 15 units of R

70/30 is like a fraction and N is in the numerator

Can you mix insulins in the same syringe? yes- Clear to Cloudy- draw up the R first and then the N- RN's do it that way

Which vial do you inject the air into first? (pressurizing the vial)- inject air into N first, then inject the air into the R, then draw up the R and then draw up the N – NRRN- always ends in RN

What needle will you use for giving a particular injection?

If they say IM- and which needle you'll use?

a) 21 gauge- 5/8 inch
b) 22 gauge- 1 inch
c) 21 guage-1 inch
d) 25 gauge- 5/8 inch

Pick the answer in which both parts have a 1 in it so C would be the answer- what if they changed it to Subq

S- looks like 5- pick the answer that has a 5 in both parts so the answer will be D

Heparin Vs. Coumadin

Heparin is given IV or Subq
Coumadin is given only PO
Heparin works immediately
Coumadin takes a few days to a week to work
Heparin cannot be given for longer than 3 weeks (except for lovenox)- in 21 days you start making heparin antibodies-
Coumadin can be taken forever
Antidote to heparin overdose is protamine sulfate – heP- Pro
Antidote to Coumadin is vitamin K- Cou K
Lab value that monitors heparin is PTT- count your fingers
Lab value that monitors Coumadin is the PT INR- count your fingers
Heparin can be given to pregnant women
Coumadin cannot be given to pregnant women

*only psychotic tranquilizer to be given to pregnant women is haldolol

K wasting and K sparing diuretic:

Any diuretic ending in the letter X- x's out K so it is a waster + diuril - lasix
The others are all sparers- spironolactone

Backlophen- muscle relaxants
2 side effects- fatigue and muscle weakness
3 things you teach: don't drink, don't drive and don't operate heavy machinery
- when you are on your backlophen you are on your back lo phen

Flexeril- other muscle relaxant

- FLEX- Muscle

SEMIDEs- are typically X's

Pediatric teaching
How would you teach children (Piaget)
4 stages for kids thinking
0-2 – sensory/ totally present oriented (don't think about the past or future)- teach them while you do it/teach them as you doing/ teach them what you are doing and verbally, tell them. Ex. Kid getting spinal lumbar puncture, while you are doing it, tell him what you are doing- for this age group you need to pre teach the parents not the kids

3-6- preoperational (Preschooler)- fantasy oriented, imaginative, illogical, they do understand the future and the past, teach them shortly before "the morning of or the day of"- don't given them enough time to tie their imagination into, teach them what you are going to do- talk in future tense, learn through play, 4 year old going to lumbar puncture- tell him the day of what you are going to use using play. Cannot teach them skills because they start imagining

7-11 year old concrete operation kids- (concrete 711's) rule oriented, live and die by the rules, rigid, one and only way to do things, "my parents said".. "my teacher said", "you're doing it wrong"- not doing it like the other person. Need to teach them days ahead- teach them what you're going to do + skills because they like to go by the rule- use age appropriate reading and demonstration

12-15 year old- formal operations- kids can abstract- can think cause and effect- once they ask questions--- always teach like an ADULT. **What is the first age a child can manage their own care?** 12, managing doesn't mean doing everything, it means they know what they can handle

Psych principles

1) Make sure you know what phase of the relationship you are in
2) Don't give gifts or accept gifts
3) Don't give advice- "what do you think you should do"
4) Immediacy- if a patient says something- do the one that keeps them talking- "lets talk about it right here and right now"- don't refer to the social worker
5) Concreteness- do not use slang- don't tell an upset patient to "chill out" because they take you literal or concrete, never use the dumb words they use either
6) Empathy- NEED empathy- best answers are those that the nurse acknowledges how the patient is reading
   a. Skip- don't worry, you shouldn't feel, most people feel
   b. Always say: that's very upsetting, that's very sad

4 step process for answering empathy questions

1) Always have a quote and each answer is a quote
2) Put yourself in the clients place
3) If I said those words and really meant them, how would I feel right now
4) Choose the answer that reflect that feeling or anything close- empathy ignores what is said and goes with what is felt

Lecture 12
Prioritization

1) Which patient is sickest or healthiest-disaster
    a. ex. in your town and you need to discharge to make room
    b. you receive report on 4 pts, which would you go to first
2) 4 parts: age, gender, diagnosis and modifying phrase (age and gender are not important but is important in peds)
3) Diagnosis is important + modifying phrase (more important)
    a. if pt had angina pectoris vs. myocardial infarction- who is the highest priority?- with unstable BP/with stable vital signs – ANGINA because he has unstable BP HIGH priority

1) Acute beats chronic- much higher priority
    a. COPD, CHF, appendicitis (highest priority because its acute)
2) Fresh Post op (12 hours) beats medical or other surgical
    a. COPD, CHF, 2 day by pass graft, 2 hour post choley (most important because fresh post op), radical neck dissection, bilateral above the knee above the knee amputation
3) Unstable beats stable-
    a. Words that count for stability: stable, chronic illness, post op greater than 12 hours, local or regional anesthesia, lab abnormalities of an A or B level, phrases ready for discharge, to be discharge or admitted longer than 24 hours ago, unchanged assessments nothing new,
    b. Words that count for instability: unstable, acute illness, post op less than 12 hours, general anesthesia in the first 12 hours, lab abnormalities of a C or a D, not ready for discharge, newly admitted, newly diagnosed or admitted less than 24 hours ago, changing or changed assessment,
    c. Patient is stable if they are experiencing the typical expected signs and symptoms of the disease with which they were diagnosed
    d. Patient is unstable if they are experiencing unexpected signs and symptoms of the disease with which they were diagnosed

Examples: who is the highest priority
-16 yr old F with meningiccocal meningitis 103.8 deg for 3 day (usually high priority because its acute, "who has had"- unchanged… puts in lower priority, high temperature expected putting her at low priority, she was also admitted less than 24 hours ago making her even lower) – she can go home with IV antibiotic
-Male with IBS who spiked a temp of 100.3 this afternoon (usually low priority because its chronic, higher priority because it's a new temperature (SPIKED) and not expected with IBS)- he cannot go home because he has a symptom that's out of the ordinary

4 things that are ALWAYS unstable
1) Hemorrhage – do not confuse with bleeding (depends on whether its expected or not)
2) HIGH fever- over 105- they will have a seizure
3) Hypoglycemia- low sugar expected or not they are in trouble
4) Pulselessness or breathlessness (at the scene of an unwitnessed accident they are the lowest priority because they are dead but if you witness the accident they are the highest priority)

3 things that result in a black tag in an unwitnessed accident
1) pulsessness
2) breathlessness
3) fixed and dilated pupils

Tie Breakers – caution only use this as a tie breaker
1) the more vital the organ the higher the priority – is it the organ of modifying phrase is happening not the diagnosis itself
2) organ vitality- brain, lung, heart, liver, kidney, pancreas

Highest priority with K+
1) heart
2) lung
3) brain (WINS)

DELEGATION (do not delegate to LPNS)
1) Starting an IV
2) Hanging or mixing IV meds
3) Pushing IV push meds (can only maintain and document the flow)
4) Cannot administer blood or mess with central lines (NO flushing or change central line dressing)
5) Not allowed to plan care- RN does the care plan LPN can implement it
6) Cannot provide teaching but can reinforce
7) Not allowed to take care of unstable patients
8) Not allowed to do the first of ANYTHING (RN needs to)
9) Not allowed to do the following: admission, discharge, transfer, or first assessment after there has been a change (crackles in lungs)

DO not delegate to an Unlicensed aide

1) charting (can only chart what they did not about the patient
2) cannot give meds except for topical over the counter barrier creams
3) cannot do assessments except for vitals and accuchecks- brain damage
4) Cannot do treatments except for enemas – cautious of allowing them to catheterize

DO NOT DELEGATE TO THE FAMILY SAFETY RESPONSIBILITIES- you are responsible for the patient, but you can delegate to a sitter because you must teach them and write it in the chart- cant let them do it without documenting and teaching

STAFF Management – inappropriate behavior of staff

1) 4 answers usually given - tell supervisor, confront them and take over immediately, at a later date just talk to them about it, ignore it
   a. "is what they are doing illegal" if the answer is yes, then always pick tell supervisor. If not then ask yourself...
   b. is anyone in danger? If the answer is yes then confront immediately and take over, if illegal do this first and then tell supervisor. If not..
   c. " is this behavior legal, not harmful but simply inappropriate" approach them later on and talk to them

4 quadrants: where is the organ they are asking for

Point and click for auscultating the valves of the heart: aortic (2nd right intercostal space), pulmonic (2nd left intercostal space), tricuspid (4th intercostal space at the left sternal border) and the mitral (5th intercostal space in the mid clavicular line) there is no leeway

Apical pulse would be the mitral area

How do you guess?

Psych questions:

- Nurse will examine their own feelings about... "something"- that way you don't countertransfer
- Establish a trust relationship

Nutrition:

- in a tie- always pick chicken
- if chicken is not there I pick fish
- never pick casseroles for children (tuna, tofu)
- never mix medication in children's food
- always need to ask for permission to mix med in food
- toddlers always get finger food
- preschoolers- leave them alone- 1 meal a day is okay- eat when they're hungry and may eat the same thing everyday for 7 weeks

Pharmacology

- Don't memorize routes or dosages, side effects are more important
- If you know what a drug does but don't know the side effects what should you do: pick a side effect in the same body system where the drug is working
- If the drug was a GI drug: pick diarrhea, heart: pick tachycardia, CNS: drowsiness
- If you don't know what the drug is, if it is PO pick a GI side effect
- Never tell a child that medicine is candy

OB

- Check fetal heart rate

Med Surgical

- First thing you assess will always be level of consciousness NOT airway
- Then you do you ABC's- establish an AIRWAY first

Pediatric

- All based on the principle- always give the child more time (grow and develop) – do not rush their growth and development
- When in **doubt,** call it normal because that implies you are giving the child more time
- When in doubt, pick the older age in the 2 that your down to because you are giving the kid more time
- When in doubt, pick the easier task when picking between to (ex. Rolling over and sitting up)
- NORMAL, OLDER, EASIER

*Rule out absolutes, but there are some like: never give K+ IV, never give a med to a patient unless you can identify it

*If two choices are the same then it is not the answer

*If two answers are opposite, one of them are probably right

*Umbrella strategy: all of the above but its not an option- look for the umbrella answer, that covers all the others without saying it does

*If the question gives you 4 right answers and asks you to pick the one with the highest priority (in regards to patients) however, if they give you one patient and list needs always pick the one that has the worst outcome

*When you're stuck between two answers, reread the question

*SESEME street rule: right answers tend to be different than the others because it's the only one that is right, but all the wrong answers all share something in common

*Don't be tempted to answer a question based on ignorance rather than knowledge (if you don't know what something there, pull it out of the question)

*Always go with your gut unless you can prove the other is superior- and use common sense

3 expectations you should not have:

1) Don't expect 75 questions. ALWAYS expect 265
2) Don't expect to know everything because its computer adaptive
3) Don't expect everything to go right